

Particles and Nuclei 2

- 2/12/26
- Announcements:
- HW 3 will be released tonight → Due in two weeks (Feb 26)
- Reading: Primarily Griff Ch 4 and 5 (this time more emphasis on end of 4, and 5).

Outline today

- Finish up few more things about on rapidity
- More on Cross sections/Distributions
- Griff Chap 4

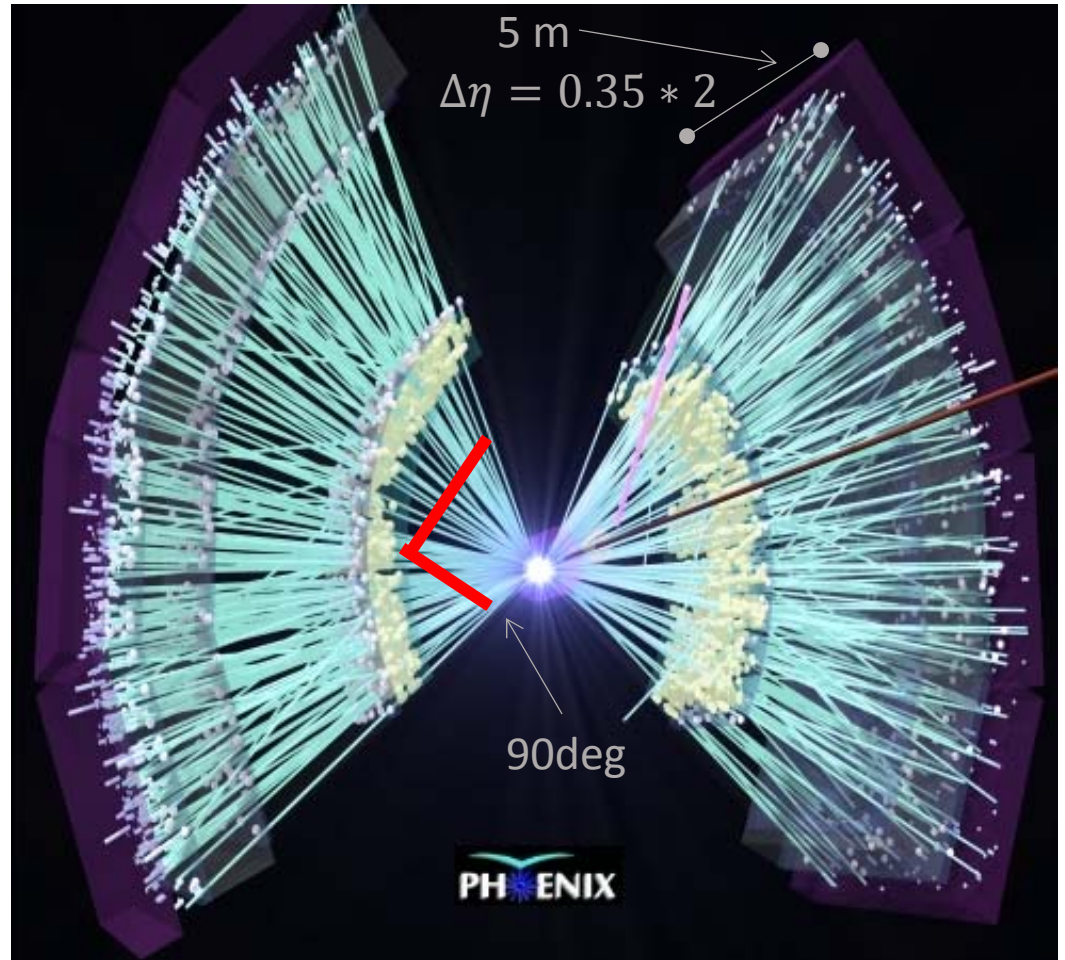
Solid angle & High Energy Hadron-Hadron Collision

- Example : PHENIX "Solid angle"
- Quoted in terms of η
- instead of $\Delta\theta$ using boost invariant $\Delta\eta \sim 2 * 0.35 * \pi$ ($\Delta\phi$)
- sPHENIX will improve rapidity acceptance to $[-1, 1]$ (2/0.7 times better) (and in ϕ π to 2π)

~ 2 sterRadian?

$\sim 1/6$ of 4π

- From areas : faces are 5 m from beam point and 5 m along beam
 - 1 arm ~ 7 m arc length 5m along beam
 - "rectangles of $5 \times 7 \text{ m} = 35 \text{ m}^2$
 - $35 \text{ m}^2 / 4\pi (5 \text{ m})^2$
 $= A_{\text{phx}} / A_{\text{full sphere}}$
 $= 2 * 35 / 270 \sim 2/9 \sim 1/5$
 fraction of total 4π solid angle a little different but close to $1/6$ above
 $\Delta\Omega = 4\pi * (1/5) = \sim 2 \text{ sR}$



Experimental Uses of Eta

- Eta basically replaced theta for many things. For example

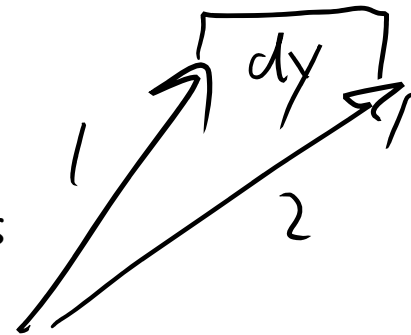
$$\vec{R} \quad \Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$$

and “ $\vec{R}$ ” = $[\eta, \phi]$ is used somewhat like solid angle -- instead of solid angle in fact acceptance is given in terms of “ $d\vec{R}$ ”, $d\phi$, Delta eta w/ (respect to beam center of detector) and elements of detector are identified by their η, ϕ locations.

Notice ΔR is “TRANSVERSE” opening angle $\rightarrow$ usefully invariant

$$\frac{d\phi}{dy} (d\eta)$$

-Recall again the arguments about y /radians



$$|d\eta| = |d\theta|$$

- From the definition of η it is easy to show that at 90 degrees (at $\eta = y = 0$) $d\eta = d\theta$:

$$\eta = -\frac{1}{2} \ln \left(\tan \frac{\theta}{2} \right)$$

$$\frac{d\eta}{d\theta} = -\frac{1}{2} \frac{1}{\tan \theta / 2} \left(\frac{1}{\cos^2 \theta / 2} \right)$$

$\downarrow \quad \theta = 90^\circ$
 $\quad \quad (\frac{\theta}{2} = 45^\circ)$

$$d\eta = -d\theta$$

$\downarrow = 1/1/2$

- This means the scale of $d\eta$ or $\Delta\eta$ is in radians (there @ 90)
- By the boost invariance of $d\eta = dy$, this means its scale is equivalent to radians at ALL y !!!!

Example Jets

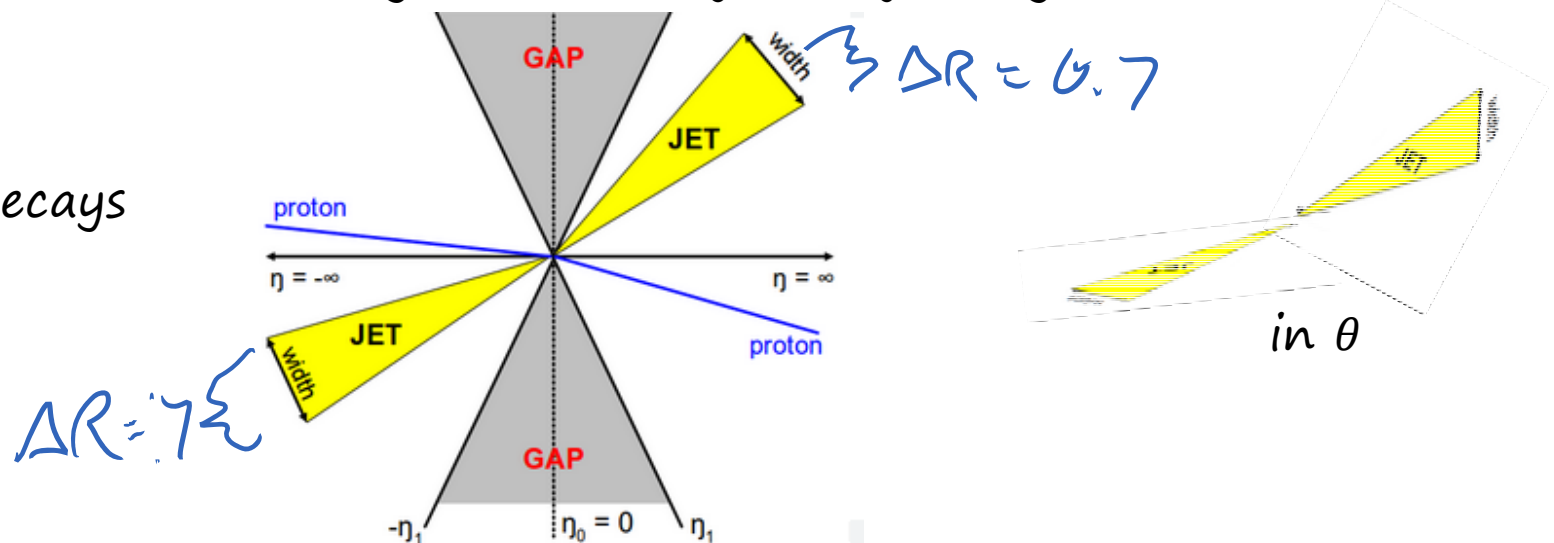
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$$\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$$

and “ $\vec{R}$ ” = $[\eta, \phi]$ is used somewhat like solid angle -- instead of solid angle in fact acceptance is given in terms of “ $\Delta\vec{R}$ ”, $\Delta\phi$, Delta eta w/ (respect to beam center of detector) and elements of detector are identified by their η, ϕ locations

-R is used a lot when talking about QCD jets $\rightarrow$ jet angular size
 $\Delta R \sim 0.7$

-Jets and Decays



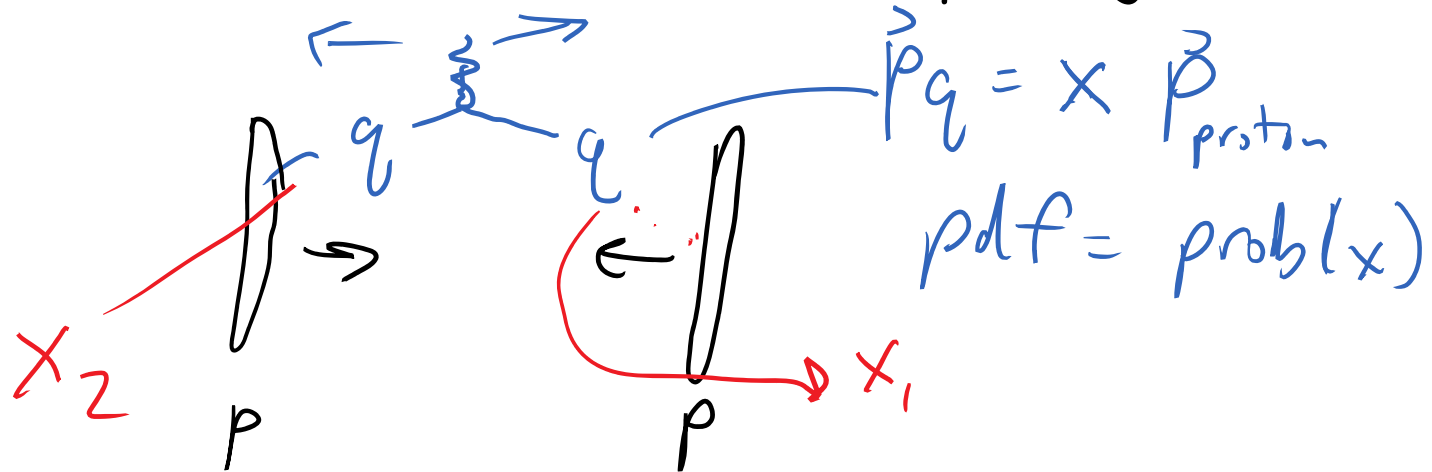
Rapidity Especially Important Hard $h+X$ collisions

- Especially important for hadrons ($h+h$, $e+h$...)
- Parton $x \rightarrow$ Fraction of mom of hadron "quark/gluon" has

Recall

p_{CM}

- Drawing.



$$\hat{S} = x_1 x_2 S$$

x_1 vs x_2 determines boost of parton COM (p_{CM})

So especially important because "beam" / Fixed Target vs collider is changing event by event depending on x_1 , x_2 (random event to event)

Turns out higher p_T (transverse momentum) final states most likely for $x_1 = x_2$ (COM frame back in lab)

General Considerations → Collider Detectors & Relativistic HI collisions

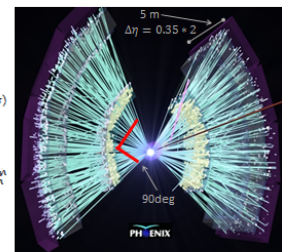
- $x_1=x_2$ means high pt distribution very center (η or $y=0$) peaked—most high pt acceptance there
- Trigger detectors near beam → beam rapidity
- Generally $\Delta\eta$ between two particles being greater than ~ 3 means not likely correlated

e.g. 2 particles from decay

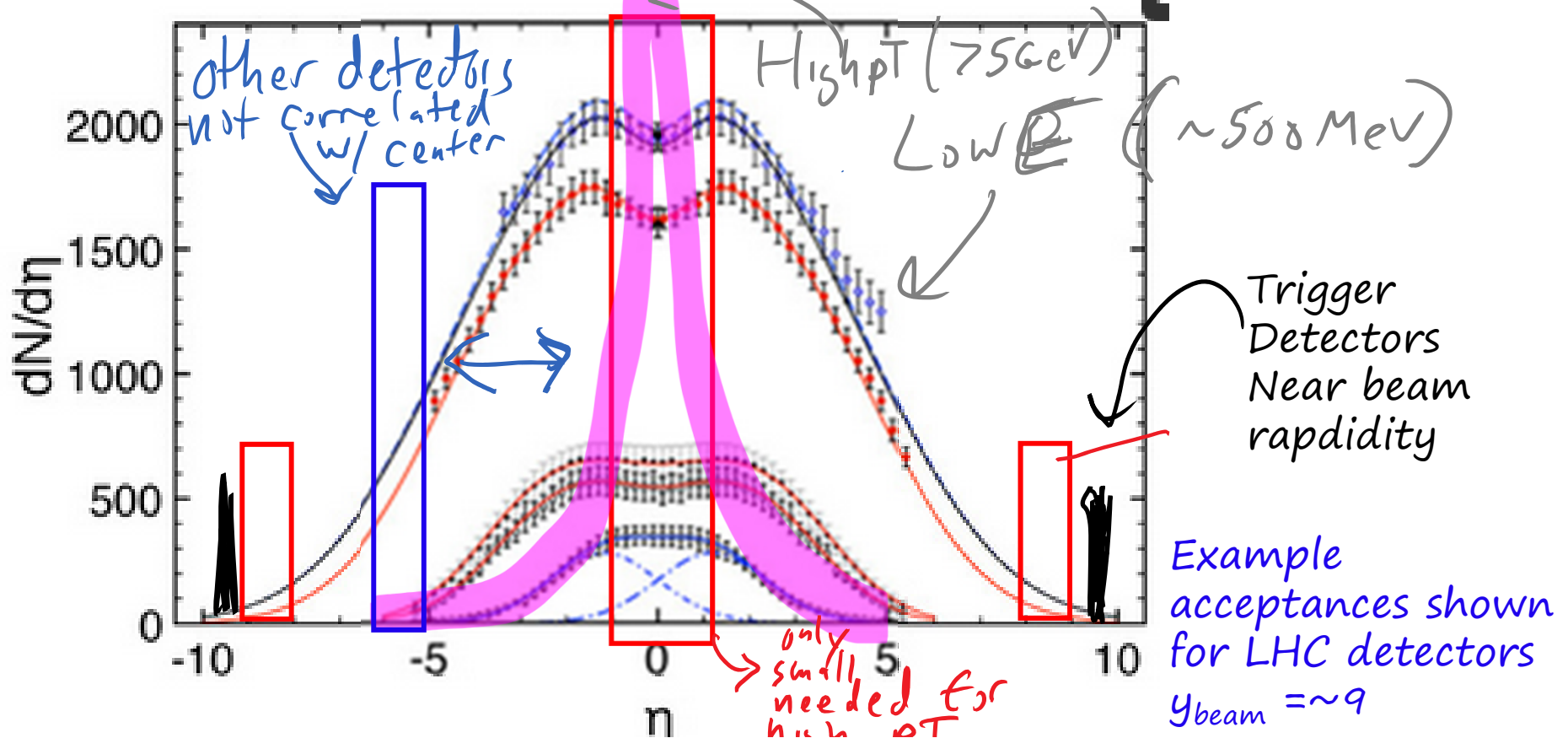
“3 radians $\sim$ whole decay phase space

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- sPHENIX will improve rapidity acceptance to $[-5, 5]$ (2/0.7 times better) (and in ϕ to 2π)
- ~ 2 steradian?
- $\sim 3/6$ of 4π



- From areas: faces are 5 m from beam point and 5 m along beam
- 1 arm ~ 7 m arc length 5 m
- “rectangles of $5 \times 7 \text{ m}^2 = 35 \text{ m}^2$ ”
- $35 \text{ m}^2 / 4\pi (5 \text{ m})^2 = 0.44$
- $\sim 2 \times 35 / 4\pi \sim 2.75$ fraction of total 4π solid angle a little different but close to 2/3 above
- $\Delta\Omega = 4\pi (1/5) \sim 2$ sr



η Hyperbolic Relations

- It's best to use these hyperbolic relations to use η to determine things like p_T

$$p_z = p_T \sinh \eta, \quad |\mathbf{p}| = p_T \cosh \eta. \quad (\cos \theta = \tanh \eta !)$$

- (Note normally:

- $p_T = p \sin \theta$
- $p_z = p \cos \theta$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

Cosh, Sinh, Tanh y – The other rapidity

- Obviously $e^{2y} = (1+\beta)/(1-\beta)$ $y = \frac{1}{2} \ln \left(\frac{1+\beta}{1-\beta} \right)$ $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$
- It shouldn't be too surprising that $\tanh y = \beta$ (if F' or else, = " $\beta_{||}$ ")
 - $f(x) = 1-x/(1+x)$ is inverse of itself (over $[0,1]$)—"involution" (nearly so elsewhere)
 - this fact is useful for some other cases in particle physics especially "asymmetries" $\rightarrow (x_1-x_2)/(x_1+x_2)$
- Further, for the situation 1 case, : $\gamma = \cosh y$ and $\gamma\beta = \sinh y$ – convenient e.g. for coding
- For this reason, sometimes rapidity is first introduced as the "hyperbolic angle" of a boost—like an alt way of specifying β and γ .

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh w & -\sinh w \\ -\sinh w & \cosh w \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} = \Lambda(w) \begin{pmatrix} ct \\ x \end{pmatrix} .$$

- But note this is a different kind of rapidity – often a different symbol is used like w or ξ –situation 1, but used in a different way
- ...

The other rapidity: not as useful

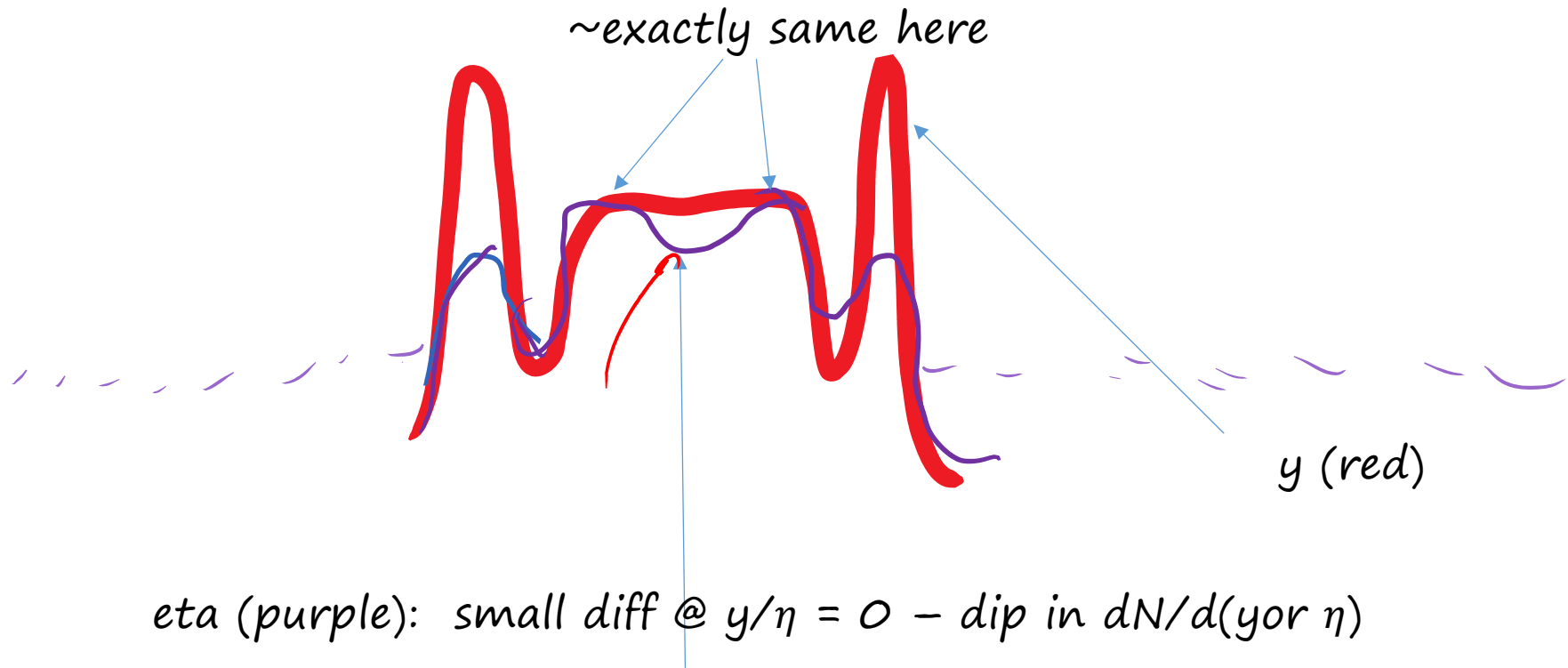
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 $y = \frac{1}{2} \ln \left(\frac{1+\beta}{1-\beta} \right)$ $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$
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- Further, for the situation 1 case, : $\gamma = \cosh y$ and $\gamma\beta = \sinh y$
- But note this is a different kind of rapidity – often a different symbol is used like w or ξ –situation 1, but used in a different way
- ...because it gives the impression that you might define different “rapidities” for each particle produced in a collision in the situation 1 way, for ALL particles ie as boosts in each particles direction



- but these “rapidities” in different directions would not have the useful transformation property (invariance of Δy 's)
 - i.e. don't think of y as boost “angle”/ boost amount for each particle
- Nonetheless the $\tanh y = \beta_{||}$ formula (situation 2) is still useful sometimes

Pseudorapidity η vs Rapidity y

- Differences from y vs η



Obviously “MAX” pseudo rapidity defined by “max” rapidity ie beam rapidity (not really max but only e.g. decays will go higher in practice) since beam pseudo-rap $\rightarrow \ln(p - p_z) = \ln(p_z - p_z) = \ln(0) = \infty$

Jacobian - Exercise

- to see the difference (next problem set) btw y and η
- Calc Jacobian $d\eta/dy$ show that it equals β
 - AT CONSTANT P_T !!!!

Start with:

$$p_z = p_T \sinh \eta, \quad |\mathbf{p}| = p_T \cosh \eta.$$

Differentiate : $dp_z/d\eta$ ($= p$)

Recall expression from previous homework $dy = dp_z/E$

Other forms

- $d^3N/dp^3 \rightarrow \equiv dN^3/dp_x p_y p_z$

- From rapidity pset $dy = dp_z/E$ (actually it was supposed to be stated in problem 3)

$$E \frac{d^3N}{dp^3} = E \frac{1}{p_T} \frac{dN}{dp_z dp_T d\phi} = \frac{1}{p_T} \frac{dN}{dy dp_T d\phi}$$

- Experimentally Common

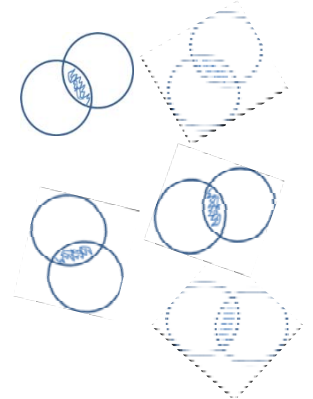
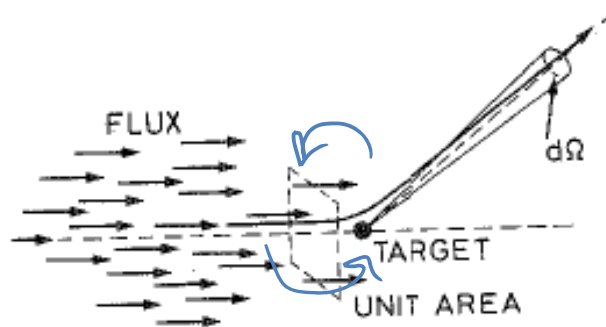
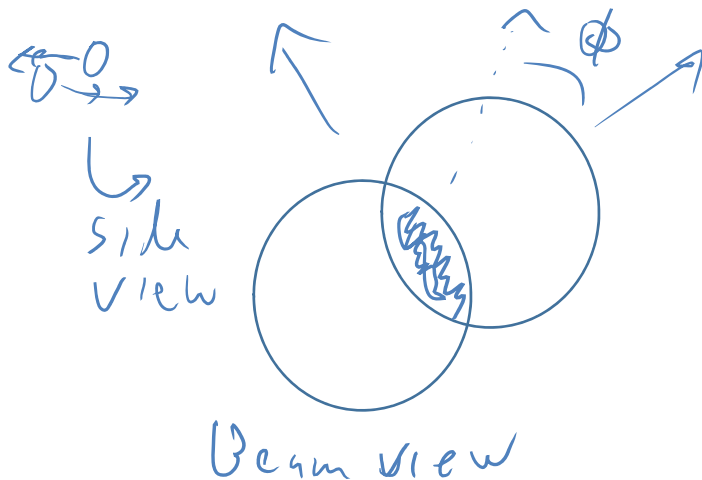
$$\approx \frac{1}{p_T} \frac{dN}{dp_T d\phi d\eta} \rightarrow \int d\phi \rightarrow \frac{1}{2\pi p_T} \frac{dN}{dp_T d\eta}$$

$\int d\eta$
 (spherical)
 dN/dp_T

- (because "phi orientation" can't be controlled ~all cross sections usually integrated over phi (N.B. not for $d\Delta\phi_{12}$ for 2 particles))

Why no ϕ dependence in scattering

- why scattering amplitude is usually just $f(\theta)$
- Most potentials e.g. (approximate or exact) point particles will have ϕ - symmetric $V(x)$ -- by symmetry
- But with beams, even if particle-particle interaction does have ϕ dependence, e.g. nuclei on left below...
 - can't control orientation of interactions and so beam will average any ϕ effect to 0



Other forms

- $d^3N/dp^3 \rightarrow \equiv dN^3/dp_x p_y p_z$

- From pset $dy = dp_z/E$

$$E \frac{d^3N}{dp^3} = E \frac{1}{p_T} \frac{dN}{dp_z dp_T d\phi} = \frac{1}{p_T} \frac{dN}{dy dp_T d\phi}$$

- Experimentally Common

$$\approx \frac{1}{p_T} \frac{dN}{dp_T d\phi d\eta} \rightarrow \int d\phi \rightarrow \frac{1}{2\pi p_T} \frac{dN}{dp_T d\eta}$$

$\int d\eta$
 (spherical)
 dN/dp_T

- (because “phi orientation” can’t be controlled ~all cross sections usually integrated over phi (N.B. not for $d\Delta\phi_{12}$ for 2 particles)

Relation between "Invariant" and Normal Phase Space

- $d^3N/dp^3 \rightarrow \equiv dN^3/dp_x p_y p_z$
- The dp^3 is called the phase space (or dp^4)
- "Available states to fill" $\rightarrow$ available directions and magnitudes
- so $dp^3 \rightarrow dp^3 / E$ not very different.
- The key part that is different is $dp_z/E \equiv dp_{||}/E$ needs to be invariant
 - p_T and ϕ (or p_x and p_y) being transverse are automatically invariant

$$\frac{dp_z}{E} dp_x dp_y = p_T \frac{dp_z}{E} dp_T d\phi$$

- Similar : recall invariant line elements

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad \text{cross product}$$

$$dp_\mu^2 \rightarrow |dp_\mu|_{4v} dp_\mu \text{ (as 4v components)}$$

$$d\vec{p}_0 \times d\vec{p}_1 \times \dots = \text{volume} \rightarrow d^4p \text{ is also invariant ! } dE dp_x dp_y dp_z$$

but E contains redundant info

Phase Space: used in relation to # states/particles

not my slides, I will find reference/citation

The state of a single-particle with momentum between 0 and p , confined to a volume V , is specified by a point in 6D phase space (x, y, z, p_x, p_y, p_z)

The extent to which spatial and momentum coordinates along each axis can be simultaneously specified is limited by the uncertainty principle, to h , so the "elemental volume" of phase space is h^3 .

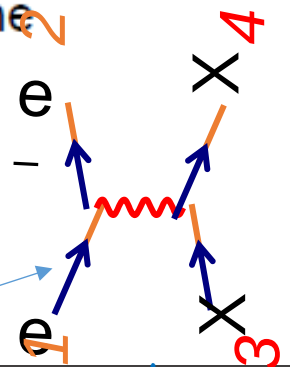
The number of states available to a single particle is thus

$$N = \frac{\text{total phase space available}}{\text{elemental volume}} = \frac{1}{(2\pi\hbar)^3} \int dx dy dz dp_x dp_y dp_z = \frac{V}{(2\pi\hbar)^3} \int d^3\vec{p}$$

assumption: "free particles" phase space "density" 1 particle per unit 6D volume— actually $1/(2\pi\hbar)^3$

For a system of n particles, the number of available final states is the product of the individual factors for each particle:

$$N_n = \left[\frac{V}{(2\pi\hbar)^3} \right]^n \int \prod_{i=1}^n d^3\vec{p}_i$$



This is all usual non-relativistic QM

3,4 for final state phase space

Particles and Nuclei 2

- 2/17/26
- Announcements:
- HW 3 Due ~Friday 2/27/26
- Reading: Griffiths 4-5 (for today) Griffiths 6.2 & 6.3, new L&B to be posted (incl Ch 11-12 mostly available w/ Ch 6/7)

List of Kinematics Topics

II. PLEASE REMEMBER

(masses are usually 0!)

A. $p \equiv (E, \vec{p})$ is a 4-vector

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$D. \left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

E. $\beta = p/E = \tanh y$

$$F. \quad E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$$

G. Lorentz transforms additive in rapidity

Relation between "Invariant" and Normal Phase Space

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$$\underbrace{dp_z}_{\equiv E} dp_x dp_y = p_T \underbrace{dp_z}_{\equiv E} dp_T d\phi$$

- Similar : recall invariant line elements

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad \text{cross product}$$

$$dp_\mu^2 \rightarrow |dp_\mu|_{4v} dp_\mu \text{ (as 4v components)}$$

$d\vec{p}_0 \times d\vec{p}_1 \times \dots = \text{volume} \rightarrow d^4p$ is also invariant ! $dE dp_x dp_y dp_z$
but E contains redundant info

Fermi's Golden Rule : Preview Ch 6

transition probability $W = \frac{2\pi}{\hbar} |\mathcal{M}_{if}|^2 \rho_f(E)$

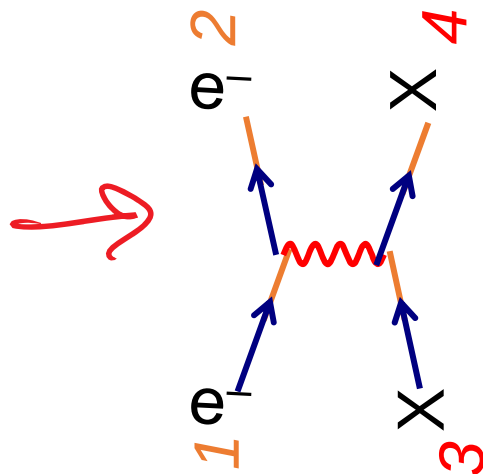
Density of final states available for energy E .

$\propto 1/(2\pi\hbar)^3$ for free particles

matrix element $|\mathcal{M}_{if}|^2 = |\langle \psi_f | V_{if} | \psi_i \rangle|^2$

Matrix element contains the fundamental physics (e.g. the dynamics). *see next slide*

- Recall : The Matrix Element comes from (Time Dependent) Perturbation Theory
- Wave Functions are of the solved $H \rightarrow H_0$
- V is the "new" "Interaction" Hamiltonian/Potential H_{int}



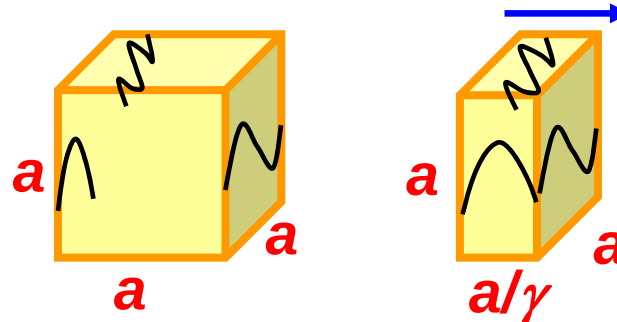
We wish to transition between
Using Free particle states $H_0 = T$

$$\langle e_1 e_2 | \quad + \quad | x_3 x_4 \rangle$$

The "propagator" field(s) is/are the perturbation
Higher order diagrams naturally higher order PT

Lorentz Invariant Phase Space “LIPS”

- In non-relativistic QM normalise to one particle/unit volume: $\int \psi^* \psi dV = 1$
- When considering relativistic effects, volume contracts by $\gamma = E/m$



- Particle density therefore increases by $\gamma = E/m$
 - ★ Conclude that a relativistic invariant wave-function normalisation needs to be proportional to E particles per unit volume
- Usual convention: **Normalise to $2E$ particles/unit volume** $\int \psi'^* \psi' dV = 2E$
- Previously used ψ normalised to 1 particle per unit volume $\int \psi^* \psi dV = 1$
- Hence $\psi' = (2E)^{1/2} \psi$ is normalised to $2E$ per unit volume

ANOTHER WAY TO SEE PROBLEM W/ NEGATIVE ENERGIES $\rightarrow$ NEGATIVE PROBABILITIES

QFT: Proof of Needing Antiparticles #2 of 2

- Griffiths: K.G. : Leads to negative probabilities AND negative energy solutions
- Negative energy solutions more OK $\rightarrow$ anti-particles Dirac, (Proca? spin 1 – see Lancaster/Bludell Massive Photon Fields section) has too
- Negative probability a problem – Griffiths: but not if we make $\Psi^*\Psi = \text{charge density}$ instead of number/counting probability
- That's not what ultimately happens though. See L/B
Instead, because the field equations have BOTH anti/particle terms Prob density/current FOR SCALAR (real) ϕ has equal ANTIPARTICLE/PARTICLE contributions

The resulting covariant probability current for the Klein-Gordon equation is then given by

$$j^\mu(x) = i \{ \phi^*(x) \partial^\mu \phi(x) - [\partial^\mu \phi^*(x)] \phi(x) \}, \quad (6.13)$$

which, as the notation suggests, is a four-vector. Substituting in our wave function $\phi(x) = N e^{-ip \cdot x}$ gives the time-like component of the probability current as

The problem $\rightarrow j^0 = \rho = 2|N|^2 E. \quad (6.14)$

Since E can be positive or negative we can't interpret ρ as a probability density, since there's no such thing as negative probability! This seems to ring the death knell for the Klein-Gordon equation as a half-decent single-particle quantum mechanical equation.

Another explanation (same result)

Consider a particle in an interaction to be at rest w.r.t. a box of volume V containing it and only it. Now consider this system viewed in a reference frame S' in which the box is moving with relative velocity v . In this frame (in which we want to calculate W) the volume V is Lorentz contracted by a factor of $\gamma = E/mc^2$ (assuming the particle has mass).

$$V' = \frac{mc^2}{E} V \quad \int \psi^* \psi dV' = \frac{E}{mc^2} \quad (\text{in } S')$$

e.g. the particle density has increased by a factor of γ . To ensure a reference-frame independent particle density normalization need to incorporate a factor of $\sqrt{2E}$ in the wave-function normalization:

$$\psi = \sqrt{\frac{2E}{V}} e^{-i(\vec{p} \cdot \vec{r} - Et)}$$

This wave-function normalization is Lorentz invariant.

*This version explicitly shows the $1/V$ still
this is a classical wave fn but the QFT expressions will still contain
these elements plus other “modifiers”/ operators (e.g. a $a^\dagger$)*

OLD Wave Functions

- Since now we will be talking about free particle fields [QFT] with multiparticles: Typically normalize WF to a density of 1 per unit volume V ie. $1/V$

$$\psi = \sqrt{\frac{2E}{V}} e^{-i(\vec{p} \cdot \vec{r} - Et)} \quad \int \psi^* \psi dV' = \frac{E}{mc^2} \quad (\text{in } S')$$

- In Quantum Field Theory these V (=Volumes) crop up a lot in certain derivations $\rightarrow$ e.g. Ch 4 L&B
- Volume integrals over "ALL SPACE" = V
- I take it as a reflection that we are always describing scattering of particles/"potentials" active over very small but macroscopic space V around beam crossing point/target. So usual "bound" state-ing integral "infinity" away is not very far $\rightarrow$ like theta function

$$V \rightarrow \boxed{Vol} \rightarrow r \text{ or } z$$

In Other words for Griff Ch. 6

transition probability $W = \frac{2\pi}{\hbar} |\mathcal{M}_{if}|^2 \rho_f(E)$  Density of final states available for energy E .

matrix element $|\mathcal{M}_{if}|^2 = \left| \langle \psi_f | V_{if} | \psi_i \rangle \right|^2$ Matrix element contains the fundamental physics (e.g. the dynamics).

- We don't change Fermis Rule, but by multiplying by $1 = E/E$
- We make the phase space density /integral $d^3p/E \rightarrow$ invariant (also equals $= d^2p_T dy!!!!$)
- and the Wave Function normalizations invariant
- The other factors in the various Golden Rule Expressions in Ch 6 come from further details of imposing Lorentz Invariance
- Except statistical factor (later other e.g. "color" factors)

Griffiths Ch 6 Golden Rules

- Griffiths prefers to use d^4p but later changes...

$$\Gamma = \frac{S}{2\hbar m_1} \int |\mathcal{M}|^2 (2\pi)^4 \delta^4(p_1 - p_2 - p_3 \cdots - p_n) \\ \times \prod_{j=2}^n 2\pi \delta(p_j^2 - m_j^2 c^2) \theta(p_j^0) \frac{d^4 p_j}{(2\pi)^4}$$

- L&B Explains the connection uses LIPS d^3p/E

Example

- For the two body decay

$$i \rightarrow 1 + 2$$

$$\begin{aligned} M_{fi} &= \langle \psi'_1 \psi'_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i \cdot 2E_1 \cdot 2E_2)^{1/2} \langle \psi_1 \psi_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i \cdot 2E_1 \cdot 2E_2)^{1/2} T_{fi} \end{aligned}$$

- ★ Now expressing T_{fi} in terms of M_{fi} gives

$$\Gamma_{fi} = \frac{(2\pi)^4}{2E_i} \int |M_{fi}|^2 \delta(E_i - E_1 - E_2) \delta^3(\vec{p}_i - \vec{p}_1 - \vec{p}_2) \frac{d^3 \vec{p}_1}{(2\pi)^3 2E_1} \frac{d^3 \vec{p}_2}{(2\pi)^3 2E_2}$$

Note:

M_{fi} uses relativistically normalised wave-functions. It is **Lorentz Invariant**

$\frac{d^3 \vec{p}}{(2\pi)^3 2E}$ is the **Lorentz Invariant Phase Space** for each final state particle
the factor of $2E$ arises from the wave-function normalisation
(prove this in Question 2)

This form of Γ_{fi} is simply a rearrangement of the original equation
but the **integral** is now **frame independent** (i.e. **L.I.**)

Γ_{fi} is inversely proportional to E_i , the energy of the decaying particle. This is
exactly what one would expect from time dilation ($E_i = \gamma m$).

Energy and momentum conservation in the delta functions

Master Golden Rules

Golden Rule for Decays

Statistical Factor

Consider decays of the form $1 \rightarrow 2+3+ \dots + N$

$$d\Gamma = |\mathcal{M}|^2 \frac{S}{2\hbar m_1} \left[\left(\frac{cd^3\vec{p}_2}{(2\pi)^3 2E_2} \right) \left(\frac{cd^3\vec{p}_3}{(2\pi)^3 2E_3} \right) \dots \left(\frac{cd^3\vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 - p_2 - \dots - p_n)$$

- For scattering

For the process $1 + 2 \rightarrow 3 + 4 + \dots + N$

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3\vec{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3\vec{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3\vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \dots - p_n)$$

Master Golden Rules

- For scattering (cross section $d\sigma$) :

For the process $1 + 2 \rightarrow 3 + 4 + \dots + N$

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3\vec{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3\vec{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3\vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \dots - p_n)$$

$$\int \frac{d\sigma}{dp_3^3} = \text{X}$$

So often we will be calculating the same kind of invariant x.s.'s meaning as function of say $\vec{p}_3$ [by "dividing" to the left side by d^3p_3/E_3], by integrating over the inv. phase space of all the other final state 4-momenta,

Griffiths Chapter 4

- Clebsch Gordon Coefficients – That's especially what I want to skip
- So far you've used them to combine quantized angular momentum states.
- These are really just an extension of the (complex-number) linear algebra of “tensor products” – could work for any finite dimensional vector spaces
- “Group Theory (in Physics?)” : Just “tricks” for making higher dimensional vector spaces out of tensor products of lower dimensional ones ?
- As we stated before, the vector elements we want to manipulate represent various properties of the SM particles (e.g. “flavor”, charges, ang momentum, helicity, chirality others...)

Charges

- (QED) Electric charge 2 charges $+$ $-$ “U(1) algebra”
 - magnitude units $|e| \rightarrow g_{em} \rightarrow \alpha$
- (QCD) Color charge 6 charges $r, g, b, \bar{r}, \bar{g}, \bar{b}$ SU(3)
 - - neutralize two ways color-anticolor = “white” all three colors/anti's $rgb = \text{white}$
 - “magnitude” $g_s \rightarrow \alpha_s$
 - In QFT Represented by SU(3) matrices that satisfy this algebra (See below)
- Best to see by what is the “color” of the gluons:

 <https://slideplayer.com/slide/8741005/>

6 of the gluon fields are independent linear combinations of the simple gluon fields we enumerated

$$\begin{aligned} G_\mu^1 &= (r\bar{g} + g\bar{r})/\sqrt{2} & G_\mu^4 &= (b\bar{g} + g\bar{b})/\sqrt{2} & G_\mu^6 &= (r\bar{b} + b\bar{r})/\sqrt{2} \\ G_\mu^2 &= i(r\bar{g} - g\bar{r})/\sqrt{2} & G_\mu^5 &= i(b\bar{g} - g\bar{b})/\sqrt{2} & G_\mu^7 &= i(r\bar{b} - b\bar{r})/\sqrt{2} \end{aligned}$$

pg 285
of
Griffiths

These are 9D/3D vectors $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ i \\ 0 \end{pmatrix}$

Group Theory $\rightarrow$ Charges

- Mathematical way to make these properties so that they can be combined in a way that won't lose some aspect of their separateness
 - as opposed to scalar charge if just plus/minus will cancel out—spins can combine but usually the each “spinning” particle remains distinct.
 - combining “orthogonal” properties in linear combinations for satisfying new quantum operators
 - “properties” $\rightarrow$ quantum states/ kets
- FLAVOR “u vs d”-ness “p vs n”-ness “approximately also conserved”
- Chiral Symmetry - Chirality/Helicity ChPT χ EFT (Spontaneously Broken)
 - P.S.: - RHI $\rightarrow$ Chiral symmetry “restoration”
- Isospin - “Longer range” lower E - Parity, G-Parity, CP, CPT
- Even Electric Charge becomes more complex than $+/-$:
 - L&B : Ch 7/12 $\rightarrow$ U(1) symmetry
 - NEXT SLIDE

Symmetries – Noethers Theorem

Every symmetry of nature is associated to a conservation law, and *vice versa*.



symmetries

- ✓ Translations in time → conservation of energy
- ✓ Translations in space → conservation of linear momentum
- ✓ Rotations in space → conservation of angular momentum
- ✓ Gauge transformations → conservation of charge

e.g. $\psi \rightarrow e^{i\alpha} \psi$

Here ψ is the electron wavefunction.

Gauge transformation is “global” if $\alpha = \text{const}$, “local” if $\alpha = \alpha(x)$

charges

Our fundamental theories (e.g. those forming the Standard Model) are all based on the principle of local gauge invariance.

NOT all “charges” have “forces/fields” associated with them
Isospin is one of those (appropriately named?)
Its conservation

L&B Complex Scalar Field

- This is the simplest way EM charge is modeled in QFT

68 Examples of Lagrangians, or how to write down a theory

7.6 The complex scalar field

We can also make a transformation that simplifies the two-scalar-fields Lagrangian considered in the previous section. We define two new fields ψ and $\psi^\dagger$ given by

$$\begin{aligned}\psi &= \frac{1}{\sqrt{2}}[\phi_1 + i\phi_2], \\ \psi^\dagger &= \frac{1}{\sqrt{2}}[\phi_1 - i\phi_2].\end{aligned}\quad (7.16)$$

This transformation can be pictured by an Argand diagram with ϕ_1 along the real axis and ϕ_2 along the imaginary axis [Fig. 7.3(a)]. Using the new fields we obtain a new Lagrangian

$$\mathcal{L} = \partial^\mu \psi^\dagger \partial_\mu \psi - m^2 \psi^\dagger \psi - g(\psi^\dagger \psi)^2. \quad (7.17)$$

This is known as a complex scalar field theory. Although we originally made it up from two sorts of field, we can imagine that it describes one sort of complex-valued field ψ . This new field is constructed from a bit of ϕ_1 and a bit of ϕ_2 and retains two degrees of freedom. The new Lagrangian (eqn 7.16) is invariant with respect to rotations in the complex plane [Fig. 7.3(b)]

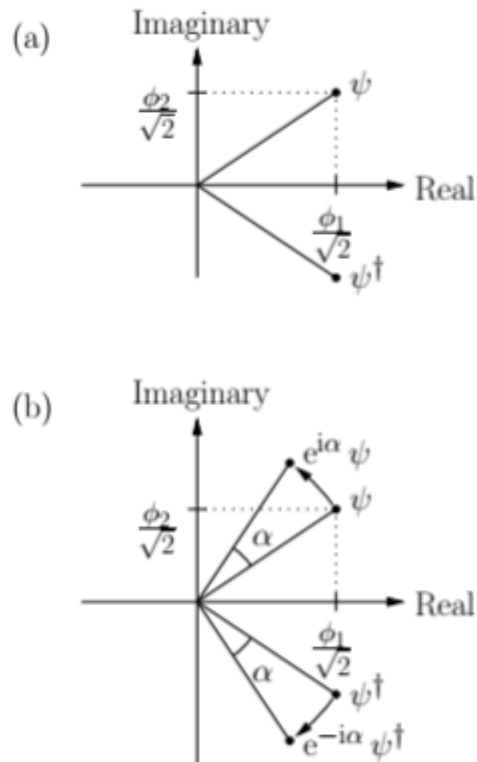


Fig. 7.3 The complex scalar field on an Argand diagram.

L&B Complex Scalar Field

- Already alluded to in L&B Ch 6:

Noether's theorem: $U(1)$ internal symmetry

$$\begin{aligned} D\psi &= i\psi & D\psi^\dagger &= -i\psi^\dagger \\ \Pi_\psi^\mu &= \partial^\mu \psi^\dagger & \Pi_{\psi^\dagger}^\mu &= \partial^\mu \psi \\ D\mathcal{L} &= 0 & W^\mu &= 0 \\ J_N^\mu &= i[(\partial^\mu \psi^\dagger)\psi - (\partial^\mu \psi)\psi^\dagger] \\ \hat{Q}_{Nc} &= -N[\hat{Q}_N] \\ &= \int d^3p (\hat{n}_p^{(a)} - \hat{n}_p^{(b)}). \end{aligned}$$

QFT : A “procedure”
for applying Noethers
theorem

³With a subscript c for conventional, just like conventional current direction in circuit theory!

Finally, we normal order to give a conserved charge of

$$N[\hat{Q}_N] = \int d^3p (\hat{b}_p^\dagger \hat{b}_p - \hat{a}_p^\dagger \hat{a}_p). \quad (12.14)$$

The conserved charge is given by the difference in the number of antiparticles $n_p^{(b)}$ and the number of particles $n_p^{(a)}$. This reason for this is quite simple. Particles, which are excitations in the field, carry Noether charge of one sign. To have a conserved charge we need other sorts of particle excitation carrying negative charge to cancel out the contribution from the particles. We are left with the conclusion that the reason for the existence of antiparticles is that charge wouldn't be conserved if they didn't exist.

Obviously, if a current J_N^μ is conserved, then so is $-J_N^\mu$ and so our assignment of which excitations carry positive Noether charge and which are negative is quite arbitrary. It is only when we attach some observable property, such as electric charge q , to each that we decide which is which. By convention, the **number current** $\hat{J}_{Nc}^\mu$ is defined to be positive for particles (created by $\hat{a}_p^\dagger$ operators) and negative for antiparticles (created by $\hat{b}_p^\dagger$ operators) and so we will define⁶ $\hat{J}_{Nc}^\mu = -N[\hat{J}_N^\mu]$, $\hat{Q}_{Nc} = \int d^3x \hat{J}_{Nc}^0$ and therefore $\hat{Q}_{Nc} = -N[\hat{Q}_N]$. Hence using eqn 12.14 and recalling that $\hat{n}_p^{(a)} = \hat{a}_p^\dagger \hat{a}_p$ and $\hat{n}_p^{(b)} = \hat{b}_p^\dagger \hat{b}_p$, we can conclude that

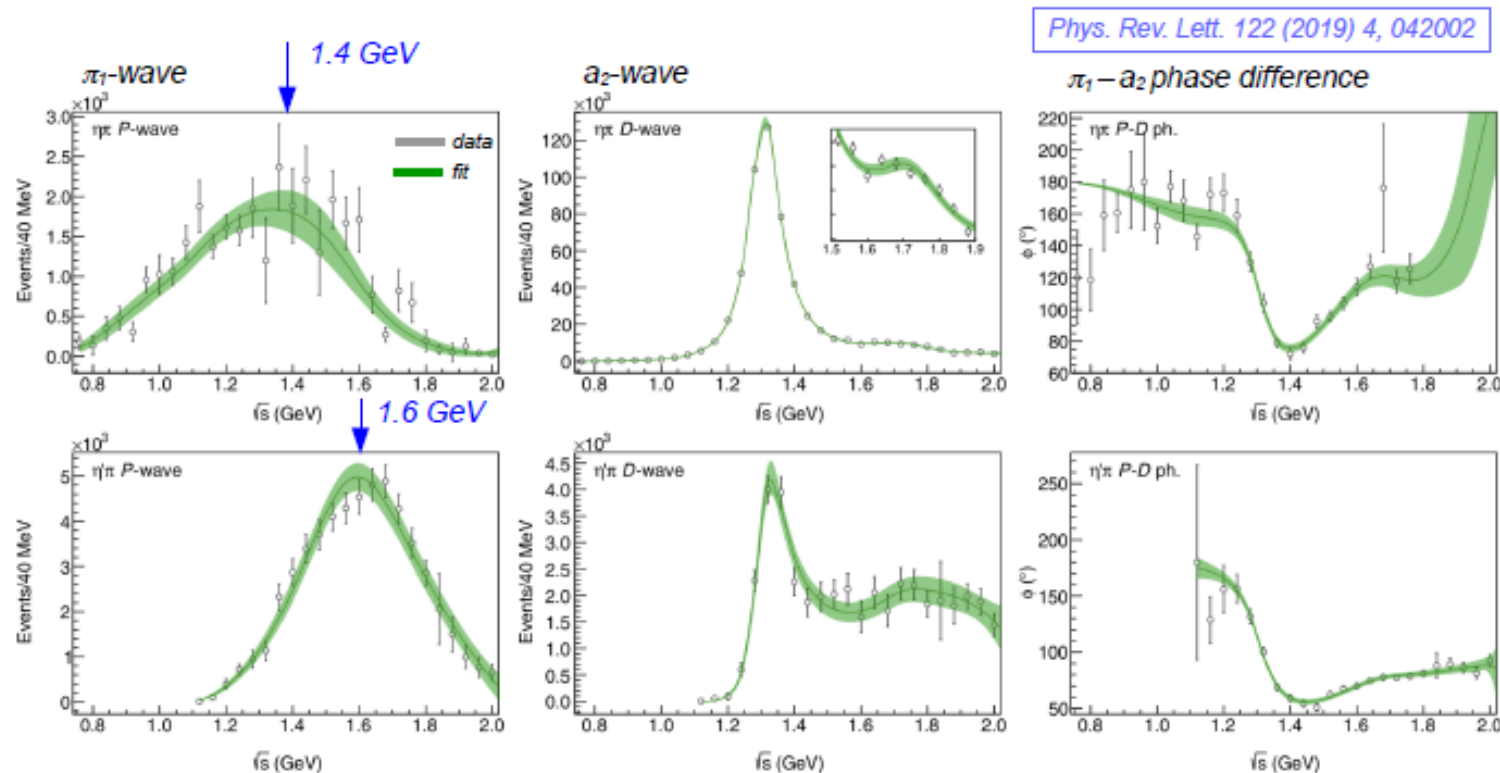
$$\hat{Q}_{Nc} = \int d^3p (\hat{n}_p^{(a)} - \hat{n}_p^{(b)}). \quad (12.15)$$

Symmetries – Experimentally

- Experimentally we're not using these much / phenomenology
- Some cases where it's interesting to know many of these rules
 - E.g. how many pions or photons a meson can decay to
- ...But in practice we can just use the PDG to “discover” or predict these things
- Relevant if you want to look for (or you find!) a new resonance peak
- But this is its own subfield I should mention I've heard seminars on some Jlab experiments (CLAS – Djali?!?–Hicks)
- P.W.A
 - Complex method of statistical subtractions of cross sections over wide angles/momenta (phase space) using constraints from these kind of spectroscopic properties (e.g. disentangling angular distributions) – can allow you to find certain resonances that would be buried otherwise.
 - P, C, G – P, J, L, S all built into such a model.

Partial Wave Analysis & “Coupled Channel”

- Nucl. Lunch paper
- Sensitivity to energy, angular ($\cos\theta$), mass, to disentangle different channel contributions
- Coupled Channel also different decay modes (below $\pi + \eta$ & $\pi + \eta'$)
- BIG Simultaneous Fit including symmetry (spin/isospin etc) constraints over mass and other distributions



Isospin

- These slides are from Krieger – U Toronto

Pion-Nuclon Scattering Amplitudes

$$\pi^+ + p: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad \frac{3}{2} \right\rangle$$

$$\pi^0 + p: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^- + p: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^+ + n: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^0 + n: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^- + n: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad -\frac{3}{2} \right\rangle$$

See Perkins Introduction to High Energy Physics section on "Isospin in the pion nucleon system" for more detail. (Section number varies from edition to edition; it is 3.13 in the 4th edition)

$$(a) \pi^+ + p \rightarrow \pi^+ + p$$

$$(f) \pi^- + n \rightarrow \pi^- + n$$

} Both are pure $I = 3/2$ on either side of the reaction. Both proceed purely via the $I = 3/2$ amplitude $\mathcal{M}_3$

- I, I_3 not I_3 , $m_3 z \rightarrow s, m_z$
- ($I_3 \equiv "m_{3z}"$ from last thurs)
- At start I made an analogy to isospin being a new type of charge but I also claimed it was just flavor which isn't really a "charge"?
- But it is a charge for residual nuclear strong force interactions:
- Chiral EFT explicit isospin terms in Lagrangian $\rightarrow$ effective potentials, isospin dependent

Isospin - quarks

$$u = \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle \quad d = \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle \quad s = |0 \ 0\rangle$$

Other quark flavours (c,b,t) are also isospin singlets, like the strange quark

Mesons containing (only) u,d quarks can therefore have isospin 0 or 1

$$\begin{aligned} \pi^- &= \bar{u}d & \pi^0 &= u\bar{u}, d\bar{d} & \pi^+ &= u\bar{d} & \text{isospin triplet } I = 1 \\ \eta &= u\bar{u}, d\bar{d} & & & & & \text{isospin singlet } I = 0 \end{aligned}$$

Mesons containing only one u,d quark can only have isospin = $\frac{1}{2}$

$$K^0 = u\bar{s} \quad K^+ = d\bar{s} \quad \text{form an isospin doublet}$$

$$\bar{K}^0 = \bar{u}s \quad K^- = \bar{d}s \quad \text{form a *separate* isospin doublet}$$

Isospin – Like the Deuteron case

Application example: pion-nucleon scattering

Six elastic $\pi N \rightarrow \pi N$ processes (e.g. same particles in and out of scattering)

$$(a) \pi^+ + p \rightarrow \pi^+ + p \quad (b) \pi^0 + p \rightarrow \pi^0 + p$$

$$(c) \pi^- + p \rightarrow \pi^- + p \quad (d) \pi^+ + n \rightarrow \pi^+ + n$$

$$(e) \pi^0 + n \rightarrow \pi^0 + n \quad (f) \pi^- + n \rightarrow \pi^- + n$$

and 4 charge-exchange processes

$$(g) \pi^+ + n \rightarrow \pi^0 + p \quad (h) \pi^0 + p \rightarrow \pi^+ + n$$

$$(i) \pi^0 + n \rightarrow \pi^- + p \quad (j) \pi^- + p \rightarrow \pi^0 + n$$

Pions carry $I = 1$ and nucleons, $I = 1/2$ so the total isospin can be either $1/2$ or $3/2$

Since isospin is conserved (in these strong interactions) there are two distinct (quantum-mechanical) amplitudes:

$\mathcal{M}_1$ for $I = 1/2$

Takes an isospin $1/2$ initial state
to an isospin $1/2$ final state.

and

$\mathcal{M}_3$ for $I = 3/2$

Takes an isospin $1/2$ initial state
to an isospin $3/2$ final state.

Isospin

- These slides are from Krieger – U Toronto

Pion-Nuclon Scattering Amplitudes

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$$\pi^+ + n: \quad \left| 1 \quad 1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \quad \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| \frac{1}{2} \quad \frac{1}{2} \right\rangle$$

$$\pi^0 + n: \quad \left| 1 \quad 0 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \quad -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle$$

$$\pi^- + n: \quad \left| 1 \quad -1 \right\rangle \left| \frac{1}{2} \quad -\frac{1}{2} \right\rangle = \left| \frac{3}{2} \quad -\frac{3}{2} \right\rangle$$

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} Both are pure $I = 3/2$ on either side of the reaction. Both proceed purely via the $I = 3/2$ amplitude $\mathcal{M}_3$

$$\langle 3/2 | \mathcal{M} | 3/2 \rangle \sim \mathcal{M}_3$$

What does this mean?

- Two Matrix elements :

$$M = \langle \text{final} | \hat{O} | \text{initial} \rangle$$

Conserving
is spin
const.

More info: What does this mean?

- Two Matrix elements :

$$M = \langle \text{final} | \hat{O} | \text{initial} \rangle$$

Conserving
isospin
const.

- Strong force doesn't affect isospin
 - for flavor: means (QCD interaction) doesn't depend on flavor (only on color)
 - Note this isn't entirely true for nucleons / hadronic interactions
AFAICT ie for the "residual" effective nuclear force mediated by pions
- Consider this exact case : $\hat{O}$ ($\hat{M}$?) doesn't change isospin STATE but interaction depends on I (total I , not I_3)
 - eigenvalue of $\hat{O}$ apparently, M_3 corresponds to different eigenvalues apparently compared to M_1 ?

More info: What does this mean?

- Two Matrix elements :

$$M = \langle \text{final} | \hat{O} | \text{initial} \rangle$$

Conserving
isospin
const.

- One way this works in this case e.g.

$$\pi^0 + p \rightarrow \pi^+ + p$$

$$\sqrt{\frac{2}{3}} | \frac{3}{2} \frac{1}{2} \rangle + \sqrt{\frac{1}{3}} | \frac{1}{2} \frac{1}{2} \rangle \rightarrow$$

$$M_1 = d \quad M_3 = c$$

$$a \langle 3 | + b \langle 1 | \hat{O} (a | 3 \rangle + b | 1 \rangle)$$

$$a \langle 3 | + b \langle 1 | [c a | 3 \rangle + d b | 1 \rangle]$$

$$a^2 c + b^2 d (\langle 1 | 1 \rangle)$$

Isospin

- Just slightly more complicated than the deuteron case

Pion-Nucleon Scattering Amplitudes

$$\pi^- + p: \quad |1 \ -1\rangle \left| \frac{1}{2} \ \frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2} \ -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2} \ -\frac{1}{2} \right\rangle$$

$$\pi^0 + n: \quad |1 \ 0\rangle \left| \frac{1}{2} \ -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2} \ -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} \left| \frac{1}{2} \ -\frac{1}{2} \right\rangle$$

$$(c) \pi^- + p \rightarrow \pi^- + p \quad \mathcal{M}_c = \frac{1}{3} \mathcal{M}_3 + \frac{2}{3} \mathcal{M}_1$$

$$(j) \pi^- + p \rightarrow \pi^0 + n \quad \mathcal{M}_j = \sqrt{\frac{2}{3}} \sqrt{\frac{1}{3}} \mathcal{M}_3 - \sqrt{\frac{2}{3}} \sqrt{\frac{1}{3}} \mathcal{M}_1^{em}$$

(ie. only need to consider diagonal matrix elements if interacting strongly or

em)

Cross-section σ for some process is proportional to $|\mathcal{M}|^2$

Isospin

- Just slightly more complicated than the deuteron case

$$\sigma_a : \sigma_c : \sigma_j = 9|\mathcal{M}_3|^2 : |\mathcal{M}_3 + 2\mathcal{M}_1|^2 : 2|\mathcal{M}_3 - \mathcal{M}_1|^2$$

At $E_{\text{CM}} = 1232 \text{ MeV}$ there is a dramatic bump (increase) in the pion-nucleon scattering cross-section, due to a πN resonance (the Δ). This is listed amongst the particles on the inside cover of your text

[we will discuss resonance in scattering in a future lecture]

Δ carries isospin $3/2$, therefore:

Expect the scattering at this energy to be dominated by the isospin $3/2$ channel

$$\sigma_a : \sigma_c : \sigma_j = 9 : 1 : 2$$

My continued “bashing”:

-notice again, this is LOW ENERGY scattering

-Ratios ~work mostly elsewhere, not as good though, but there is a big background under the peaks – example of inv. mass
-FRANTZ: but...the peaks are broad, and the “bkg” is still QCD – (Lattice QCD vs isospin)

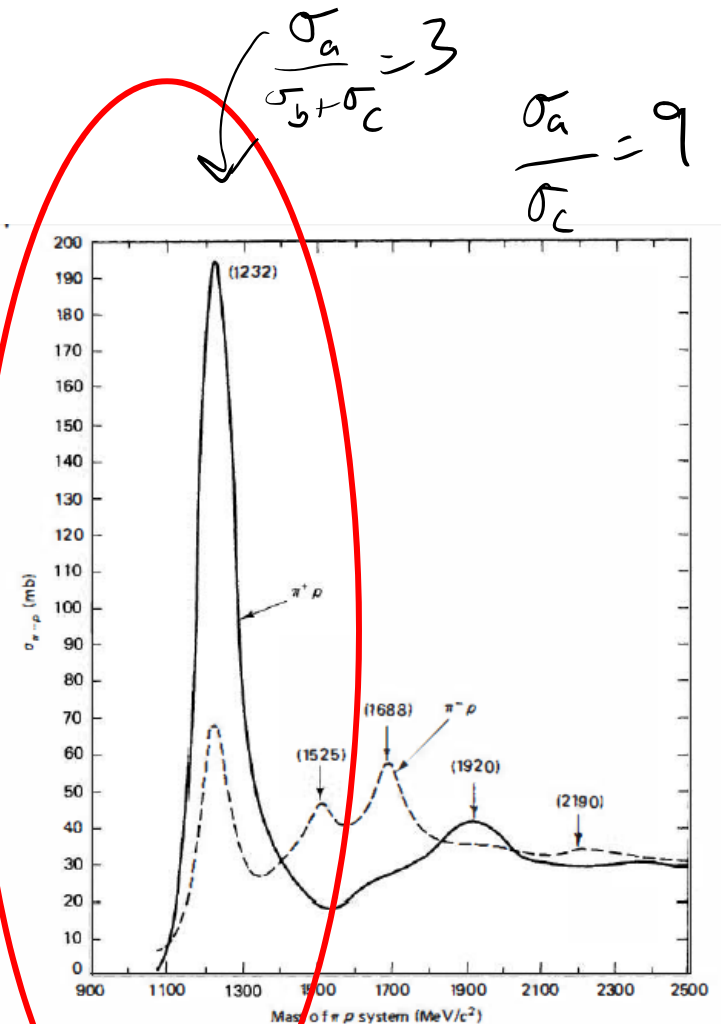


Fig. 4.6 Total cross sections for $\pi^+ p$ (solid line) and $\pi^- p$ (dashed line) scattering. (Source: Gasiorowicz, S. (1966) Elementary Particle Physics, John Wiley & Sons, New York, p. 294. Reprinted by permission of John Wiley and Sons, Inc.)

Maybe not exactly only dependent on I only

- Could even depend also a little on I_3 maybe
 - Like fine structure/hyperfine : at the level supported by data the “ $1/3$ ($1/9$?)” which isn't very exact, there could also be small differences for each I_3 state

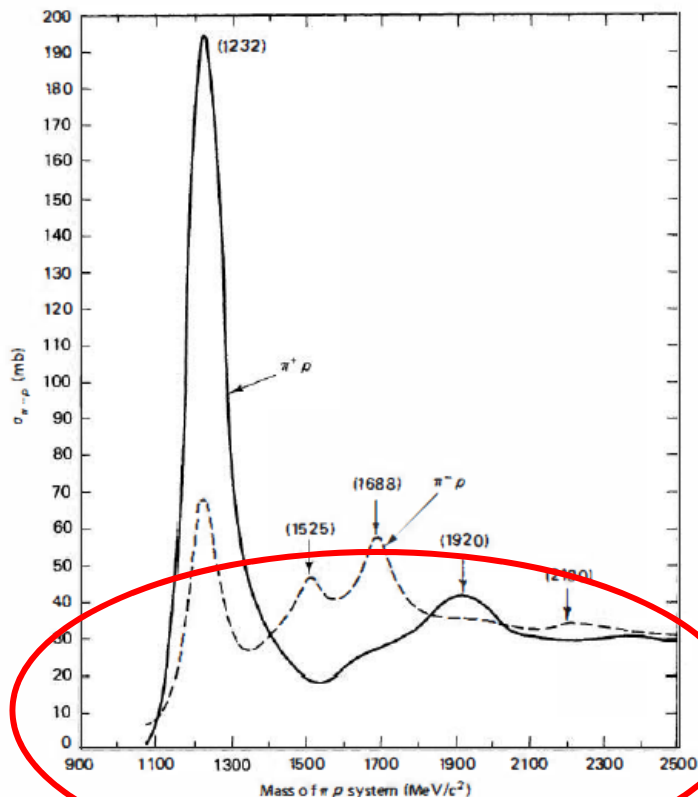


Fig. 4.6 Total cross sections for $\pi^+ p$ (solid line) and $\pi^- p$ (dashed line) scattering. (Source: Gasiorowicz, S. (1966) Elementary Particle Physics, John Wiley & Sons, New York, p. 294. Reprinted by permission of John Wiley and Sons, Inc.)

Other wise why did Heisenberg introduce the original $+1/2$, $-1/2$

(spurious extra charge aspect : I ?)

USEFUL TO RECALL

$SU(2)$ is not 2 dimensional – it's more like 4 dimensional because any element / component can always be complex

Griffiths: “ $SO(3)$ [$\mathbb{R}^3$] math structure almost identical to $SU(2)$
8 $SU(2)$ Matrix “components”
 $SO(3)$ – 9 real matrix components but probably constraints

Parity Conservation

Fundamental particles have *intrinsic* parity: $P^2 = I$ (so eigenvalues are ± 1)

Quantum Field Theory: the parity of a fermion is opposite to that of its antiparticle
the parity of a boson is the same as its antiparticle

Parity of a composite system is given by the product of the parity of the constituents, with an additional contribution of $(-1)^\ell$ according to the orbital angular momentum ℓ .

Assign positive parity to the quarks, (thus negative parity to the anti-quarks)

Mesons carry parity $(-1)^{\ell+1}$ Baryons carry parity $(+1)^3 \cdot (-1)^\ell = (-1)^\ell$

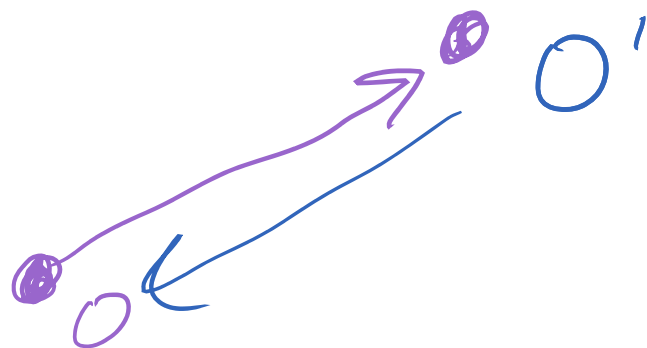
The parity of individual hadrons is one of the particle properties listed by the PDG

For example, pions have spin 0 and negative parity. They are called pseudoscalar particles.

The ρ also has negative parity. It is called a vector particle (it has spin 1).

(Parity) Charge Conj: $(-1)^{(l+s)}$

- If we exchange origins obviously $x \rightarrow -x$, $y \rightarrow -y$... = PARITY!



$$\phi = \arctan(y/x)$$

doesn't change sign

$$\theta = \arctan\left(\frac{z}{\sqrt{x^2 + y^2}}\right)$$

changes sign

The First Few Spherical Harmonics, $Y_l^m(\theta, \phi)$ ^a

$$Y_0^0 = \frac{1}{(4\pi)^{1/2}}$$

$$Y_1^1 = -\left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{i\phi}$$

$$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2} (3 \cos^2 \theta - 1)$$

$$Y_1^0 = \left(\frac{3}{4\pi}\right)^{1/2} \cos \theta$$

$$Y_1^{-1} = \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{-i\phi}$$

$$Y_2^1 = -\left(\frac{15}{8\pi}\right)^{1/2} \sin \theta \cos \theta e^{i\phi}$$

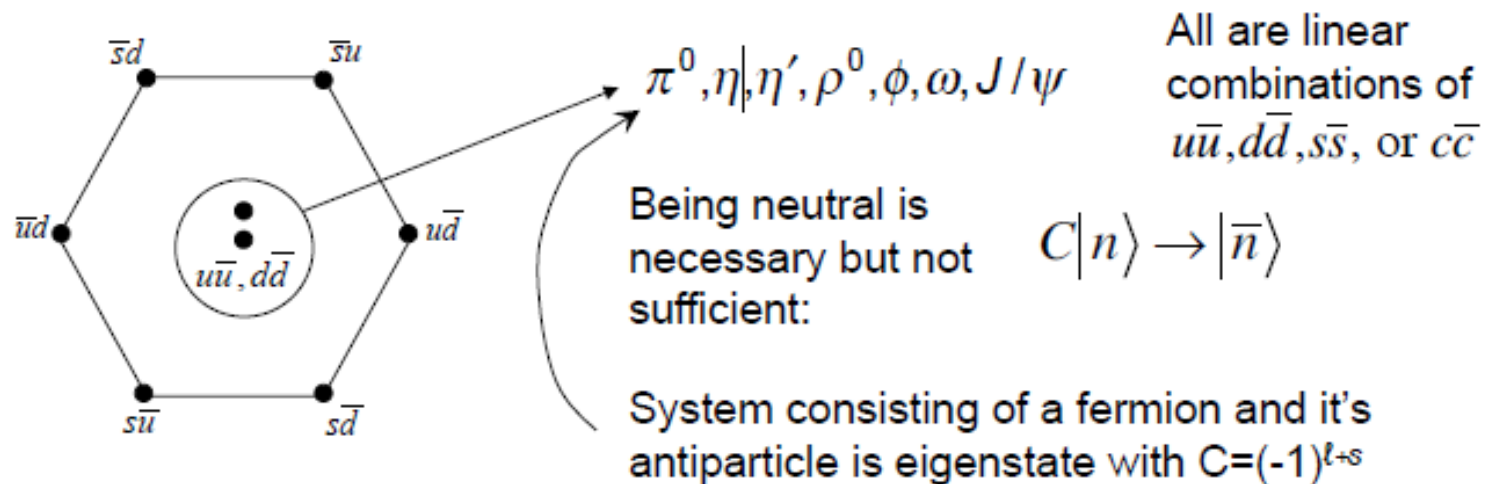
Charge Conjugation

Another symmetry operation: inverts all internal quantum numbers while leaving energy, mass, momentum, spin unchanged.

Internal quantum numbers: lepton number, baryon number, strangeness etc.

Charge conjugation takes a particle into its anti-particle.

Most particles are not eigenstates of C (particle would have to be its own antiparticle). This is true for photons, and for the central entries in the eightfold way meson nonets (made up of same-flavour $q\bar{q}$ pairs):



Charge Conjugation

- Eigenvalues -- “charge”?

$C^2 = I$ Eigenvalues are ± 1 , so for an eigenstate $|x\rangle$: $C|x\rangle \rightarrow \pm|x\rangle = |\bar{x}\rangle$

Electromagnetism is invariant under a change in the sign of all charges

The photon, γ , is the quantum of the EM field, which changes sign under C .

Photons “charge conjugation number” is therefore -1

C is a multiplicative quantum number (like parity).

System consisting of a spin $\frac{1}{2}$ particle and its anti-particle, in a configuration with orbital angular momentum ℓ is an eigenstate of C with eigenvalue $(-1)^{\ell+s}$

Invariance Under C

Strong and electromagnetic interaction are invariant under charge conjugation

Consider the electromagnetic decay of the neutral pion. The π^0 is the lightest meson and so cannot decay strongly. Instead it decays electromagnetically with branching fraction

$$\text{BR}(\pi^0 \rightarrow \gamma\gamma) = 98.8\% \quad \text{and} \quad \text{mean lifetime } 8 \times 10^{-17} \text{ s}$$

C is +1 before $[(-1)^0]$ and after $[(-1)(-1)]$ for the photon pair $[(-1)^n \text{ for } n \text{ photons}]$ so there is no decay to three photons. Similarly, $\omega \rightarrow \pi^0 + \gamma$ but never $\omega \rightarrow \pi^0 + 2\gamma$.

Other implications of charge conjugation invariance (for example): Consider process

$$p + \bar{p} \rightarrow \pi^+ + \pi^- + \pi^0 \quad (\text{strong interaction})$$

Charge conjugation invariance requires the energy distributions of the two charged pions in the final state must be equal. !!!!!

*Influences "Dynamics" again like isospin
(again note however, this is low E phenomena not
quark level)*

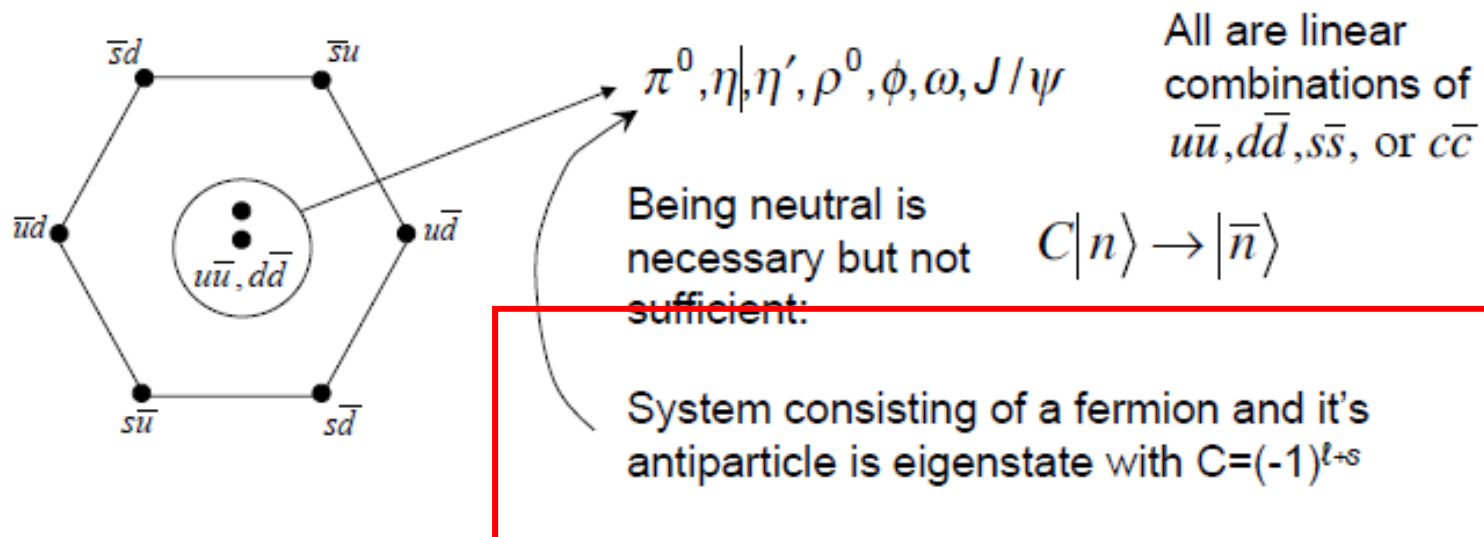
Charge Conjugation

Another symmetry operation: inverts all internal quantum numbers while leaving energy, mass, momentum, spin unchanged.

Internal quantum numbers: lepton number, baryon number, strangeness etc.

Charge conjugation takes a particle into its anti-particle.

Most particles are not eigenstates of C (particle would have to be its own antiparticle). This is true for photons, and for the central entries in the eightfold way meson nonets (made up of same-flavour $q\bar{q}$ pairs):

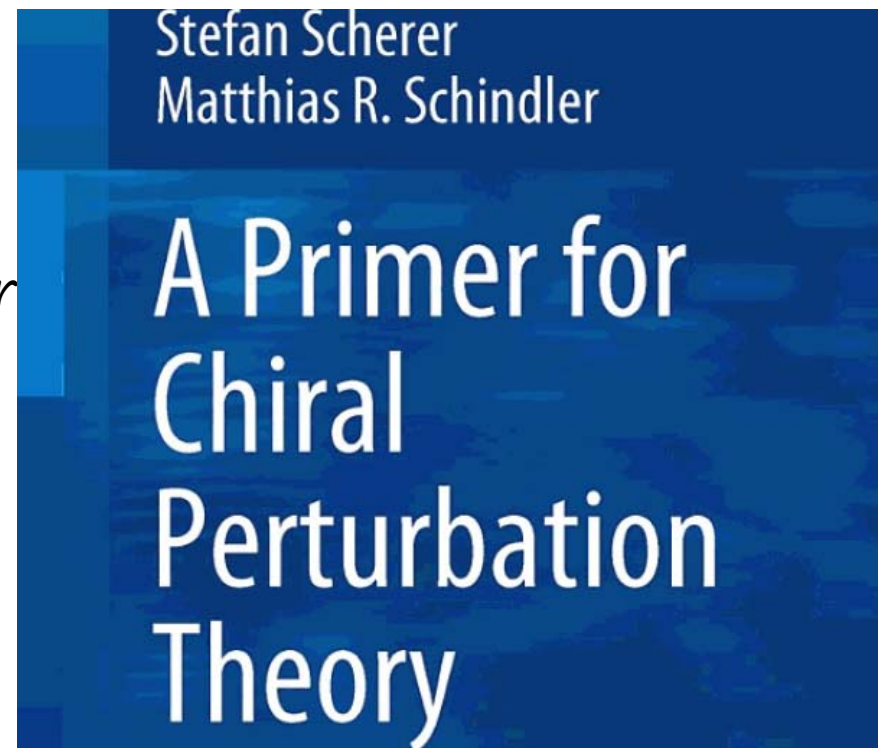


Particles and Nuclei 2

- 2/19/26
- Announcements:
- HW 3 Due Date
- Reading: For next time → Griffiths the rest of Chapter 6
→ Lancaster Blundell Chunk (Propagator)

Isospin in Chiral EFT (χ EFT)

- There are QFT's with hadron degrees of freedom (as opposed to QED, QCD, Eweak)
 - QHD $\rightarrow$ Quantum Hadron Dynamics
 - σ meson plays big role as a force mediator
 - basically combining e.g. scalar fields $\rightarrow$ vectors
 - Not very predictive
- Chiral EFT : Rigorously equivalent to QCD in certain low energy regimes
 - It CAN be Predictive
 - Predictions CAN BE equiv to QCD
 - Limited cases $\rightarrow$ but use for other
- Which symmetries does it use
- Also uses hadron degrees of freedom



Page from Ch 7 L&B

7.4 The ϕ^4 theory

How do we get particles to interact with other particles (or, equivalently, fields to interact with other fields)? Here's the simplest recipe. The ϕ^4 Lagrangian is like the scalar case, but with an extra potential term $U(\phi)$ proportional to ϕ^4 . This term makes the theory interacting. Unfortunately it also makes it unsolvable. As we'll see, we have to use a sort of perturbation theory to make predictions from this theory. The Lagrangian is⁷

$$\mathcal{L} = \frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{1}{4!} \lambda \phi^4, \quad (7.12)$$

which leads to the (not-very-memorable) equation of motion

$$(\partial^2 + m^2)\phi = -\frac{\lambda}{3!} \phi^3. \quad (7.13)$$

The constant of proportionality λ is known as the interaction strength. Clearly, when $\lambda = 0$ we return to the massive scalar field Lagrangian and the fields don't interact.

7.5 Two scalar fields

Here's another way to make particles interact. Why not have two different types of particle in the Universe, described by two fields $\phi_1(x)$ and $\phi_2(x)$? In our simple theory they have the same mass and interact with themselves and each other via a potential energy $U(\phi_1, \phi_2) = g(\phi_1^2 + \phi_2^2)^2$ where g is the interaction strength here. Notice that multiplying out the bracket gives self-interacting ϕ^4 terms and, crucially for us, a cross-term $2\phi_1^2\phi_2^2$ which forces the two types of field to interact. The resulting Lagrangian density is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_1)^2 - \frac{1}{2}m^2 \phi_1^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2}m^2 \phi_2^2 - g(\phi_1^2 + \phi_2^2)^2. \quad (7.14)$$

We notice that we get the same Lagrangian as in eqn 7.14 even if we change our definition of the fields. (We say that the Lagrangian has a symmetry.) If we transform the two fields using the mapping $\phi_1 \rightarrow \phi'_1$ and $\phi_2 \rightarrow \phi'_2$ where

$$\begin{pmatrix} \phi'_1 \\ \phi'_2 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}, \quad (7.15)$$

- Ways to combine fields to make new kinds of theories
- Lagrangian (to do subsequent steps like make Hamilt., apply Noether's theorem for conserved currents...)

The ϕ^4 Lagrangian is like the scalar case, but with an extra potential term $U(\phi)$ proportional to ϕ^4 . This term makes the theory interacting. Unfortunately it also makes it unsolvable. As we'll see, we have to use a sort of perturbation theory to make predictions from this theory. The

- Interactions $\rightarrow$ Perturbation Theory

Interactions

Chapter 4/5 (in L/B) actually discusses these interactions already and even introduces the Feynman diagram to represent it

These are perturbations on the free particle Hamiltonian

19.1 Meet some interactions

Example 19.1

L/B

(ii) The simplest self-interaction of fields is ϕ^4 (say 'phi-fourth' or 'phi-four') theory described by a Hamiltonian

$$\mathcal{H}_1(z) = \frac{\lambda}{4!} \phi(z)^4. \quad (19.3)$$

ϕ^4 self interactions are prototype for both QFT books for describing Feynman diagrams unfortunately

This is what is used for pion+pion scattering in Chiral EFT

Complex Scalar Fields – QFT

- Complex Scalar fields [not Dirac/Fermion fields] are the QFT prototypical example of making a EM-like charge :
- “U(1)” charge -- the kind that can just cancel/annihilate like EM charge
- L/B “Complex Scalar field theory section”

Step I: We start by writing down the Lagrangian we want to quantize.
The Lagrangian for complex scalar field theory is

$$\mathcal{L} = \partial^\mu \psi^\dagger(x) \partial_\mu \psi(x) - m^2 \psi^\dagger(x) \psi(x). \quad (12.1)$$

Step II: Identify the momentum density and write down the Hamiltonian. Each component² σ of the field (i.e. $\sigma = \psi$ and $\psi^\dagger$) has a different

$$+ \frac{1}{2} [\partial_\mu \phi_2(x)]^2 - \frac{1}{2} m^2 [\phi_2(x)]^2$$

We noticed that we can make a transformation that simplified this Lagrangian. We defined

$$\begin{aligned} \psi &= \frac{1}{\sqrt{2}} [\phi_1(x) + i\phi_2(x)], \\ \psi^\dagger &= \frac{1}{\sqrt{2}} [\phi_1(x) - i\phi_2(x)], \end{aligned}$$

Lagrangian

similar to two interacting real $\phi_1 \phi_2$'s – Griff's Ch 6 ABC theory (ABB see next slide) or 2 component vector fields

Also notice NOT 2 complex fields $\rightarrow$ (only 2 “components”, not 4 free “components”)

in those cases, though doesn't have this “standard” kind of “charge” “charge” here is a true conserved +/- derived from Noethers

Page from Ch 7 L&B

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Perturbation theory

- Combination $\rightarrow$ Symmetry in vector space

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- Also uses hadron degrees of freedom

Stefan Scherer

Matthias R. Schindler

A Primer for
Chiral
Perturbation
Theory

Flavor in ChiEFT

- ChiEFT implements eightfold way in similar way
 - Normal flavor symmetry

3.4.5 Application at Lowest Order: Pion-Pion Scattering

Now that the LEC F_0 has been identified with the pion-decay constant in the chiral limit, we will show how the lowest-order Lagrangian predicts the prototype of a Goldstone-boson reaction, namely, $\pi\pi$ scattering.

Exercise 3.14 Consider the Lagrangian $\mathcal{L}_2$ in the $SU(2)_L \times SU(2)_R$ sector with $r_\mu = l_\mu = 0$,

$$\mathcal{L}_2 = \frac{F^2}{4} \text{Tr}(\partial_\mu U \partial^\mu U^\dagger) + \frac{F^2}{4} \text{Tr}(\chi U^\dagger + U \chi^\dagger),$$

where

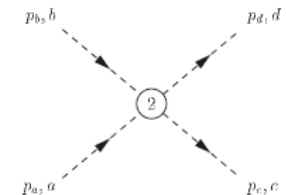
$$\chi = 2B\mathcal{M} = 2B \begin{pmatrix} \hat{m} & 0 \\ 0 & \hat{m} \end{pmatrix},$$

$$U = \exp\left(i\frac{\phi}{F}\right), \quad \phi = \sum_{i=1}^3 \phi_i \tau_i \equiv \begin{pmatrix} \pi^0 & \sqrt{2}\pi^+ \\ \sqrt{2}\pi^- & -\pi^0 \end{pmatrix}.$$

3.4 Effective Lagrangian and Power-Counting Scheme

101

Fig. 3.6 Lowest-order Feynman diagram for $\pi\pi$ scattering. The vertex is derived from $\mathcal{L}_2$, denoted by 2 in the interaction blob



by using the expansion

$$U = 1 + i\frac{\phi}{F} - \frac{1}{2}\frac{\phi^2}{F^2} - \frac{i}{6}\frac{\phi^3}{F^3} + \frac{1}{24}\frac{\phi^4}{F^4} + \dots$$

Isospin : Put in “by hand”

- *ChiEFT* still just puts isospin in “by hand”

Parity

- Pseudoscalar Mesons (Pions)
 - Spin 0
 - But Parity Operation change sign of Wave Functions
- Hmm WAVEFUNCTIONS!?!?!?

Parity Conservation

Fundamental particles have *intrinsic* parity: $P^2 = I$ (so eigenvalues are ± 1)

Quantum Field Theory: the parity of a fermion is opposite to that of its antiparticle
the parity of a boson is the same as its antiparticle

Parity of a composite system is given by the product of the parity of the constituents, with an additional contribution of $(-1)^\ell$ according to the orbital angular momentum ℓ .

Assign positive parity to the quarks, (thus negative parity to the anti-quarks)

Mesons carry parity $(-1)^{\ell+1}$ Baryons carry parity $(+1)^3 \cdot (-1)^\ell = (-1)^\ell$

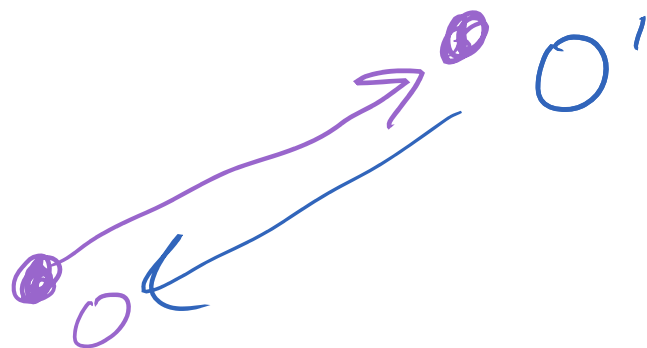
The parity of individual hadrons is one of the particle properties listed by the PDG

For example, pions have spin 0 and negative parity. They are called pseudoscalar particles.

The ρ also has negative parity. It is called a vector particle (it has spin 1).

(Parity) Charge Conj: $(-1)^{(l+s)}$

- If we exchange origins obviously $x \rightarrow -x$, $y \rightarrow -y$... = PARITY!



$$\phi = \arctan(y/x)$$

doesn't change sign

$$\theta = \arctan\left(\frac{z}{\sqrt{x^2 + y^2}}\right)$$

changes sign

The First Few Spherical Harmonics, $Y_l^m(\theta, \phi)$ ^a

$$Y_0^0 = \frac{1}{(4\pi)^{1/2}}$$

$$Y_1^1 = -\left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{i\phi}$$

$$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2} (3 \cos^2 \theta - 1)$$

$$Y_1^0 = \left(\frac{3}{4\pi}\right)^{1/2} \cos \theta$$

$$Y_1^{-1} = \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{-i\phi}$$

$$Y_2^1 = -\left(\frac{15}{8\pi}\right)^{1/2} \sin \theta \cos \theta e^{i\phi}$$

PDG

$$\pi^\pm$$

$$J^G(J^P) = 1^-(0^-)$$

$$\text{Mass } m = 139.57018 \pm 0.00035 \text{ MeV} \quad (S = 1.2)$$

$$\text{Mean life } \tau = (2.6033 \pm 0.0005) \times 10^{-8} \text{ s} \quad (S = 1.2)$$

$$c\tau = 7.8045 \text{ m}$$

$\pi^\pm \rightarrow \ell^\pm \nu \gamma$ form factors [a]

$$F_V = 0.0254 \pm 0.0017$$

$$F_A = 0.0119 \pm 0.0001$$

$$F_V \text{ slope parameter } a = 0.10 \pm 0.06$$

$$R = 0.059^{+0.009}_{-0.008}$$

π^- modes are charge conjugates of the modes below.

For decay limits to particles which are not established, see the section on Searches for Axions and Other Very Light Bosons.

π^+ DECAY MODES	Fraction (Γ_i/Γ)	Confidence level	P (MeV/c)
$\mu^+ \nu_\mu$	[b] (99.98770 ± 0.00004) %		30
$\mu^+ \nu_\mu \gamma$	[c] (2.00 ± 0.25) × 10 ⁻⁴		30
$e^+ \nu_e$	[b] (1.230 ± 0.004) × 10 ⁻⁴		70
$e^+ \nu_e \gamma$	[c] (7.39 ± 0.05) × 10 ⁻⁷		70
$e^+ \nu_e \pi^0$	(1.036 ± 0.006) × 10 ⁻⁸		4
$e^+ \nu_e e^+ e^-$	(3.2 ± 0.5) × 10 ⁻⁹		70
$e^+ \nu_e \nu \bar{\nu}$	< 5	× 10 ⁻⁶ 90%	70
Lepton Family number (LF) or Lepton number (L) violating modes			
$\mu^+ \bar{\nu}_e$	L [d] < 1.5	× 10 ⁻³ 90%	30
$\mu^+ \nu_e$	LF [d] < 8.0	× 10 ⁻³ 90%	30
$\mu^- e^+ e^+ \nu$	LF < 1.6	× 10 ⁻⁶ 90%	30

Violation of Parity

Parity Violation: the θ - τ puzzle

In the early 1950s there was an “odd” experimental observation: two particles with identical mass, spin, charge, lifetime etc, decayed (weakly) into states of opposite parity:

$$\begin{aligned}\theta &\rightarrow \pi^+ + \pi^0 & (P = +1) \\ \tau &\rightarrow \begin{cases} \pi^+ + \pi^0 + \pi^0 \\ \pi^+ + \pi^+ + \pi^- \end{cases} & (P = -1)\end{aligned}$$

Two hypotheses:

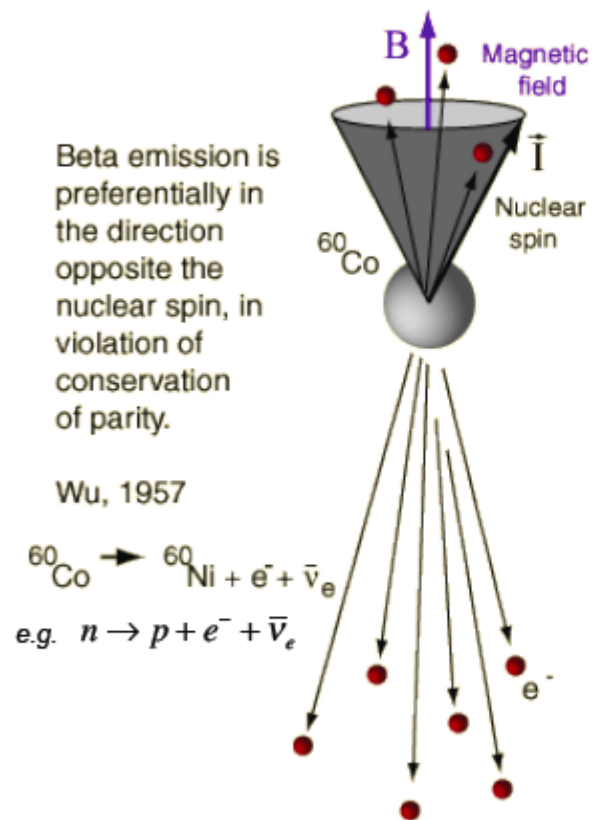
- 1) there are two particles with identical properties except for parity
- 2) parity is not conserved in weak interaction

That the weak interaction was somehow special had already been established (for instance the lifetime of “strange” particles, non-conservation of strangeness).

A survey of the literature by Lee and Yang showed there was little experimental evidence for parity conservation in weak decays.

parity violation exp

Parity Violation: Experiment (Madame Wu, 1956)



More on parity experiment

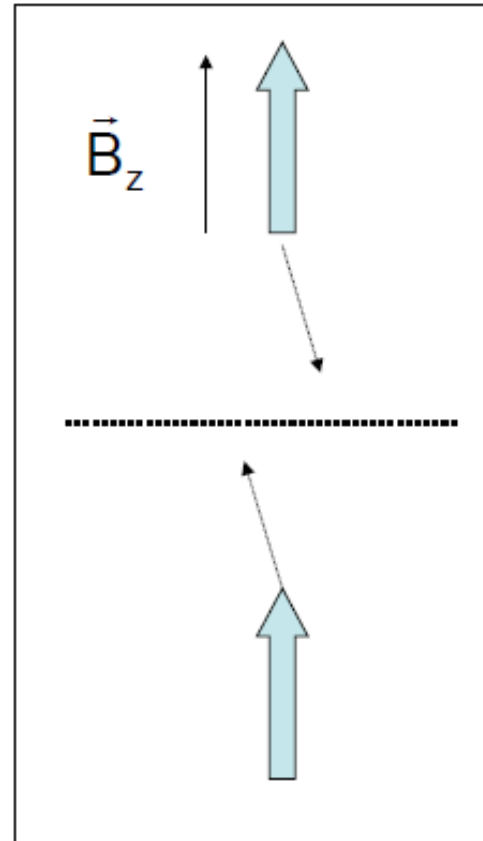
N.B. Spin is an axial vector which does NOT change sign under parity inversion.

Reflect through xy plane (here B and nuclear spin are along z).

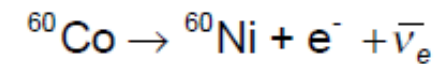
$$\vec{p}_e \rightarrow -\vec{p}_e$$

Spin does NOT reflect
(it's a pseudovector)

Parity is NOT conserved
in weak decays



^{60}Co nuclear spin



electrons preferentially
emitted in direction
opposite to nuclear spin

electrons now preferentially
emitted in direction of the
nuclear spin. This is NOT
experimentally observed.

Helicity

- Discussed before
- Use v direction as z -axis

Helicity defined as m_s/s .

Particle of spin $\frac{1}{2}$ can therefore have helicity of ± 1

Helicity +1 is referred to as right-handed



Helicity -1 is referred to as left handed.

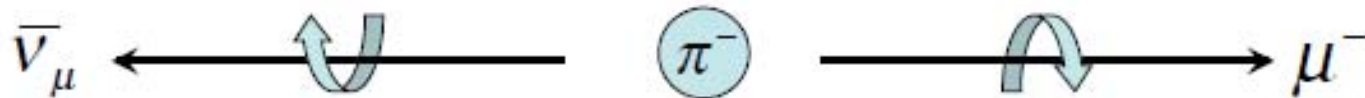


Note that helicity is NOT Lorentz invariant unless the particle is massless

If the particle has mass, one can always make Lorentz transformation into an inertial frame with velocity $> v$, and thus “flip” the helicity.

Neutrino/ Parity

Imagine the decay (at rest) of a charged pion $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$



Pion has spin 0, so spins of final state particles must be anti-aligned

Final state particles therefore have the same helicity.

A measurement of the muon helicity therefore gives a measurement of the neutrino helicity (which we have no easy way of measuring directly).

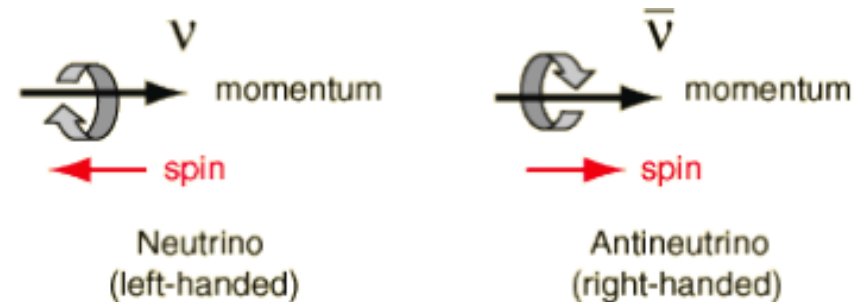
If parity were conserved, in this decay we would expect to see left-handed anti-neutrinos 50% of the time and right-handed anti-neutrinos 50% of the time.

Experimentally ONLY right-handed anti-neutrinos (e.g. as determined from the muon helicity) are observed.

actual result

All neutrinos are left-handed

All anti-neutrinos are right-handed



This absolute statement is of course not true in the case where neutrinos have mass, which we now know they do.

However, in the rest frame of the pion (as an example) it is still true that the outgoing anti-neutrino is ALWAYS right-handed.

We say that parity is maximally violated in weak decays.

(i.e. there are not simply more left-handed neutrinos than right-handed neutrinos. There are **NO** right-handed neutrinos at all.)

We will see that this parity violation is built into the dynamics of the weak interaction.

Parity operation applied to ν_L gives ν_R which does not exist.

[Momentum vector reverses but spin does not]

Exercise

- Consider ρ to 2 π decay
- $\rho \rightarrow \pi^0 \pi^0$?
- Consider from isospin and momentum exchange symmetry parity (what's parity of π^0 ?)
- ρ is vector meson quark content ud only
- ρ is isospin 1, I_3 $|1\ 0\rangle$
- Answers :
- From isospin $\pi^0 \pi^0$ not allowed because both are also $|1\ 0\rangle$ (isospin! real spin they are $|0\ 0\rangle$), meaning you could only have $|2\ 0\rangle$ or $|0\ 0\rangle$ (isospin) if 2 π^0 .
- From Parity also seems disfavored at least for strong/EM: parity of ρ -1 , parity of π^0 's also both $-1 \rightarrow$ not allowed, etc..
- From angular momentum POV, the two π^0 's would have to be in an $l > 0$ state ...possible?

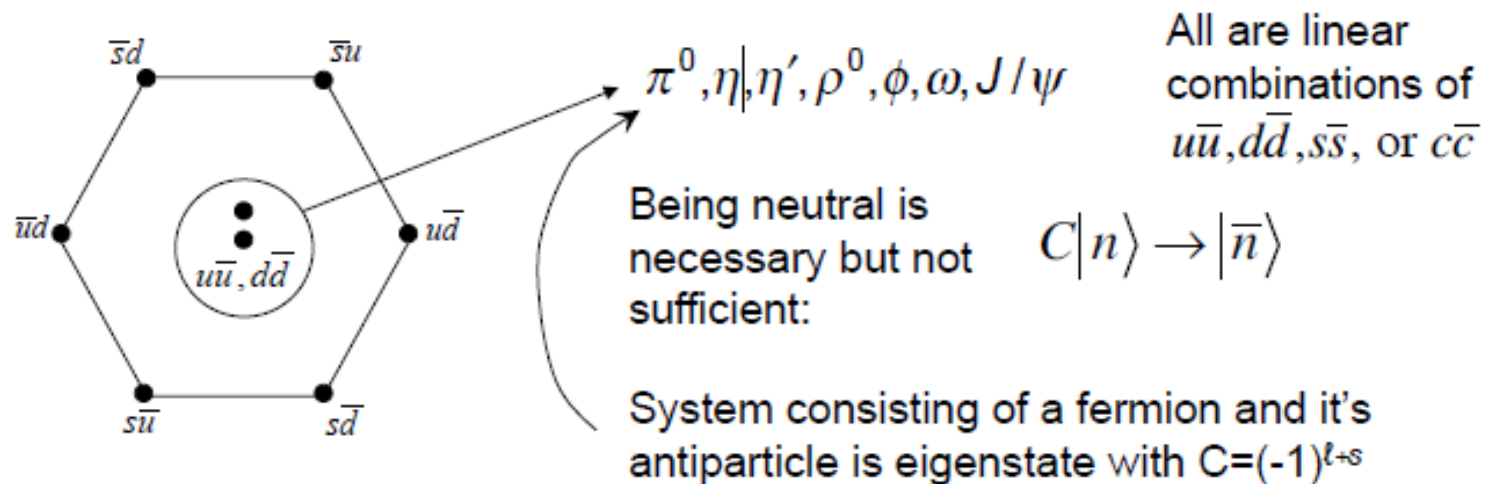
Charge Conjugation

Another symmetry operation: inverts all internal quantum numbers while leaving energy, mass, momentum, spin unchanged.

Internal quantum numbers: lepton number, baryon number, strangeness etc.

Charge conjugation takes a particle into its anti-particle.

Most particles are not eigenstates of C (particle would have to be its own antiparticle). This is true for photons, and for the central entries in the eightfold way meson nonets (made up of same-flavour $q\bar{q}$ pairs):



Charge Conjugation

- Eigenvalues -- “charge”?

$C^2 = I$ Eigenvalues are ± 1 , so for an eigenstate $|x\rangle$: $C|x\rangle \rightarrow \pm|x\rangle = |\bar{x}\rangle$

Electromagnetism is invariant under a change in the sign of all charges

The photon, γ , is the quantum of the EM field, which changes sign under C .

Photons “charge conjugation number” is therefore -1

C is a multiplicative quantum number (like parity).

System consisting of a spin $\frac{1}{2}$ particle and its anti-particle, in a configuration with orbital angular momentum ℓ is an eigenstate of C with eigenvalue $(-1)^{\ell+s}$

Invariance Under C

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Other implications of charge conjugation invariance (for example): Consider process

$$p + \bar{p} \rightarrow \pi^+ + \pi^- + \pi^0 \quad (\text{strong interaction})$$

Charge conjugation invariance requires the energy distributions of the two charged pions in the final state must be equal. !!!!!

*Influences “Dynamics” again like isospin
(again note however, this is low E phenomena not
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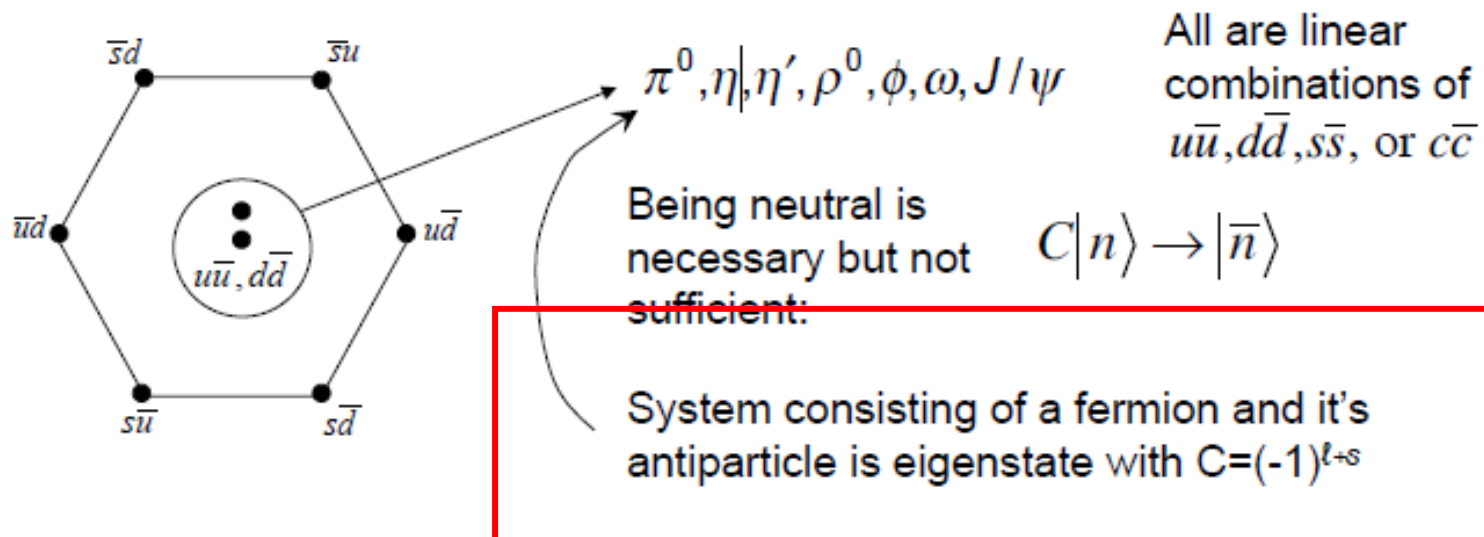
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G-parity (in strong)

Very few particles are eigenstates of the charge conjugation operator C

For strong interactions, can extend C by combining it with an isospin transformation:

Rotation of 180° about I_2 (R_2) takes I_3 into $-I_3$, for example $R_2 \pi^+ \rightarrow \pi^-$

Combining C and R_2 operations: $CR_2 \pi^+ \rightarrow \pi^+$

*nothing to do
with real rot.*

All mesons composed only of u and d quarks and anti-quarks are eigenstates of this operation which we call G or G-Parity

For particles (u,d mesons) of isospin I , the G-parity number is given by $G = (-1)^I C$

Where C is the charge conjugation number of the *neutral member* of the multiplet.

$$G(\pi) = (-1)^1(1) = -1$$

recall $C = (-1)^{l+s}$

This is a useful symmetry in strong interaction for telling how many pions can be emitted in the final state.

ρ ($I = 1$) can only decay to two pions, ω ($I = 0$) only to three)

$$G = (-1)^1(-1) = +1$$

$$G = (-1)^0(-1) = -1$$

$$\text{Summary : } G = (-1)^{\text{Isopin } I} \quad C = (-1)^I (-1)^{l+s}$$

Gparity in PDG

$$\pi^\pm$$

$$J^G(J^P) = 1^-(0^-)$$

Mass $m = 139.57018 \pm 0.00035$ MeV ($S = 1.2$)

Mean life $\tau = (2.6033 \pm 0.0005) \times 10^{-8}$ s ($S = 1.2$)

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Flavor SU(3) not Isospin

STRANGE MESONS
($S = \pm 1, C = B = 0$)
 $K^+ = us, K^0 = ds, \bar{K}^0 = \bar{d}s, K^- = \bar{u}s$, similarly for K^* 's

$K^\pm$

$$I(J^P) = \frac{1}{2}(0^-)$$

No G-parity indicated
since not relevant for
a state containing any
type of quark other
than u,d.

Mass $m = 493.677 \pm 0.016$ MeV ^[a] ($S = 2.8$)
 Mean life $\tau = (1.2380 \pm 0.0021) \times 10^{-8}$ s ($S = 1.9$)
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J^{PC} Notation

- Most common notation used for hadrons
 - comes up a lot for glueballs
 - and also “resonances” -- recall Griff “Delta Isospin” example

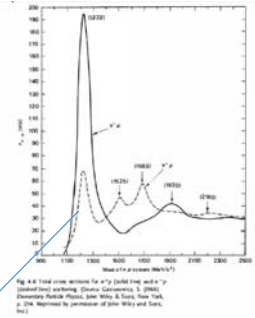


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$J^{PC} =$		$\begin{cases} 0^{-+} \\ 2^{-+} \\ \vdots \end{cases}$	$\begin{cases} 1^{+-} \\ 3^{+-} \\ \vdots \end{cases}$	$\begin{cases} 1^{--} \\ 2^{--} \\ \vdots \end{cases}$	$\begin{cases} 0^{++} \\ 1^{++} \\ \vdots \end{cases}$
$q\bar{q}$ content	$2S+1L_J =$	$^1(L \text{ even})_J$	$^1(L \text{ odd})_J$	$^3(L \text{ even})_J$	$^3(L \text{ odd})_J$
$u\bar{d}, u\bar{u} - d\bar{d}, d\bar{u}$	$(I = 1)$	π	b	ρ	a
$d\bar{d} + u\bar{u}$	$(I = 0)$	η, η'	h, h'	ω, ϕ	f, f'
and/or $s\bar{s}$					
$c\bar{c}$		η_c	h_c	ψ'	χ_c
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$t\bar{t}$		η_t	h_t	θ	χ_t

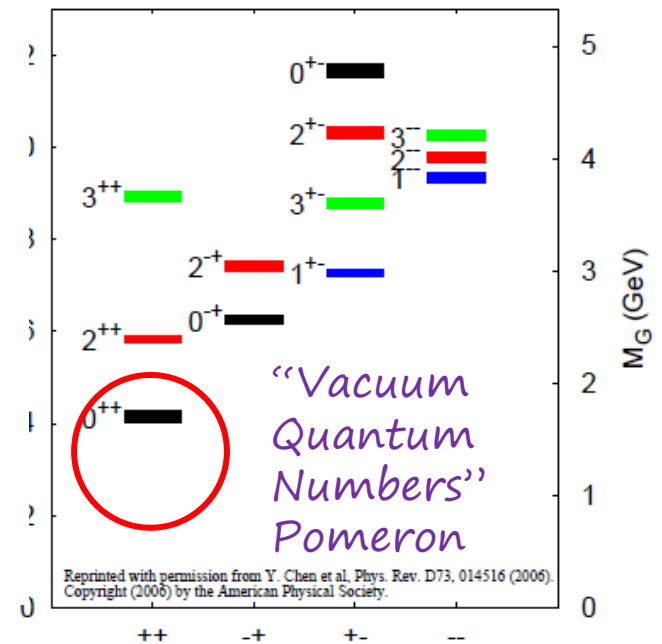


Figure 4.1: Lattice gauge prediction for pure glueball mass-th

there are lots of these
states in the pdg
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Spectroscopy <https://theory.physics.manchester.ac.uk/3.pdf>

C/CP for Weak Interaction/Strong

$$C|v_L\rangle \rightarrow |\bar{v}_L\rangle \quad \text{No!}$$

We have already stated that all anti-neutrinos are right-handed so this is an unphysical state. So charge conjugation invariance cannot be respected by the weak interaction.

Note though that the combined operations of charge conjugation and parity inversion take a left handed neutrino into a right-handed anti-neutrino

$$CP|v_L\rangle \rightarrow |\bar{v}_R\rangle \quad \checkmark$$

(the spin of the neutrino does not transform, but the velocity vector used to define the helicity does).

CP is NOT conserved by only weak –however, “just barely” not per mille effect

$$CP|v_L\rangle = |\bar{v}_R\rangle$$

*CP seems conserved in Strong Interaction
–but this may be a problem for some
→ why should it be, if electroweak is not?
STRONG CP Problem → “fine tuning” ?!?
Parity violation searched for in strong*

The way the conservation law works

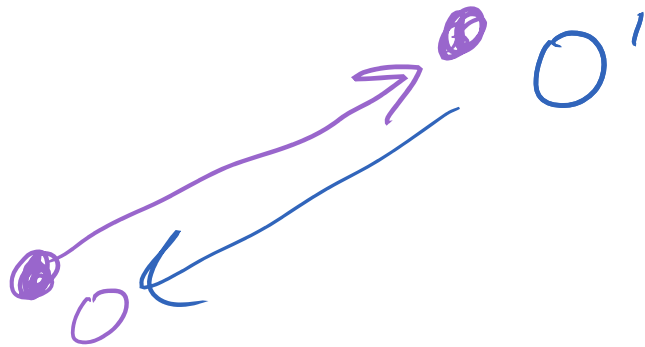
$$\begin{aligned}
 \theta^+ &\rightarrow \pi^+ + \pi^0 & (P = (-1)^2 = +1) \\
 \tau^+ &\rightarrow \left\{ \begin{array}{l} \pi^+ + \pi^0 + \pi^0 \\ \pi^+ + \pi^+ + \pi^- \end{array} \right\} & (P = (-1)^3 = -1)
 \end{aligned} \tag{4.53}$$

* This choice is completely arbitrary; we could just as well do it the other way around. Indeed, in principle we could assign positive parity to some quark flavors and negative to others. This would lead to a different set of hadronic parities, but the conservation of parity would still hold. The rule stated here is obviously the *simplest*, and it leads to the conventional assignments.

† There is less to this distinction than meets the eye; in a sense, it results from a notational anomaly. Scrupulous consistency would require that we write the parity operator in exponential form, $P = e^{i\pi K}$, with the operator K playing a role analogous to, say, spin (Equation 4.28). The eigenvalues of K would be 0 and 1, corresponding to +1 and -1 for P , and multiplication of parities would correspond to addition of K .

Charge Conj: $(-1)^{(l+s)}$

- If we exchange origins obviously $x \rightarrow -x$, $y \rightarrow -y$... = PARITY!



$$\phi = \arctan(y/x)$$

doesn't change sign

$$\theta = \arctan\left(\frac{z}{\sqrt{x^2 + y^2}}\right)$$

changes sign

The First Few Spherical Harmonics, $Y_J^m(\theta, \phi)$ ^a

$$Y_0^0 = \frac{1}{(4\pi)^{1/2}}$$

$$Y_1^0 = \left(\frac{3}{4\pi}\right)^{1/2} \cos \theta$$

$$Y_1^1 = -\left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{i\phi}$$

$$Y_1^{-1} = \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{-i\phi}$$

$$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2} (3 \cos^2 \theta - 1)$$

$$Y_2^1 = -\left(\frac{15}{8\pi}\right)^{1/2} \sin \theta \cos \theta e^{i\phi}$$

G-parity (in strong)

Very few particles are eigenstates of the charge conjugation operator C

For strong interactions, can extend C by combining it with an isospin transformation:

Rotation of 180° about I_2 (R_2) takes I_3 into $-I_3$, for example $R_2 \pi^+ \rightarrow \pi^-$

Combining C and R_2 operations: $CR_2 \pi^+ \rightarrow \pi^+$

*nothing to do
with real rot.*

All mesons composed only of u and d quarks and anti-quarks are eigenstates of this operation which we call G or G -Parity

For particles (u,d mesons) of isospin I , the G -parity number is given by $G = (-1)^I C$

Where C is the charge conjugation number of the *neutral member* of the multiplet.

$$G(\pi) = (-1)^1(1) = -1$$

recall $C = (-1)^{l+s}$

This is a useful symmetry in strong interaction for telling how many pions can be emitted in the final state.

ρ ($I = 1$) can only decay to two pions, ω ($I = 0$) only to three)

$$G = (-1)^1(-1) = +1$$

$$G = (-1)^0(-1) = -1$$

$$\text{Summary : } G = (-1)^{\text{Isopin } I} \quad C = (-1)^I (-1)^{l+s}$$

Gparity in PDG

$$\pi^\pm$$

$$J^G(J^P) = 1^-(0^-)$$

Mass $m = 139.57018 \pm 0.00035$ MeV ($S = 1.2$)

Mean life $\tau = (2.6033 \pm 0.0005) \times 10^{-8}$ s ($S = 1.2$)

$c\tau = 7.8045$ m

$\pi^\pm \rightarrow \ell^\pm \nu \gamma$ form factors [a]

$$F_V = 0.0254 \pm 0.0017$$

$$F_A = 0.0119 \pm 0.0001$$

$$F_V \text{ slope parameter } a = 0.10 \pm 0.06$$

$$R = 0.059^{+0.009}_{-0.008}$$

π^- modes are charge conjugates of the modes below.

For decay limits to particles which are not established, see the section on Searches for Axions and Other Very Light Bosons.

π^+ DECAY MODES	Fraction (Γ_i/Γ)	Confidence level	P (MeV/c)
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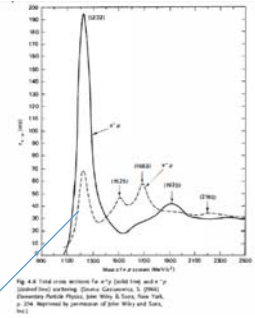


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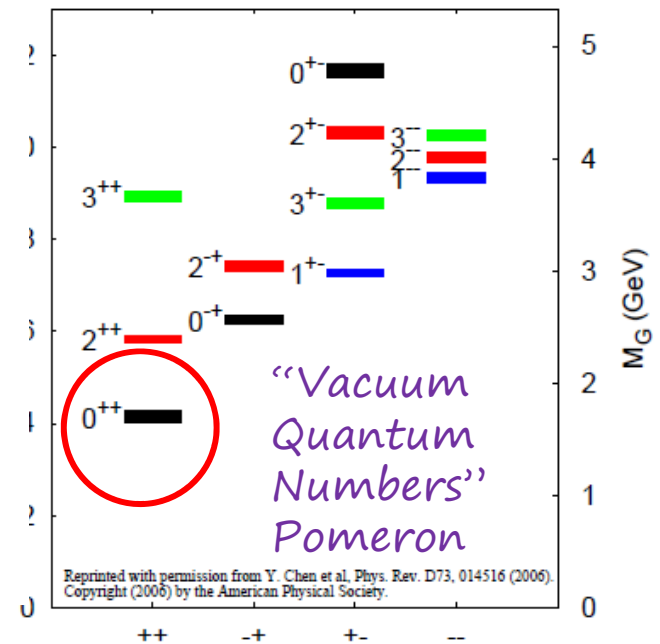


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Spectroscopy <https://theory.physics.manchester.ac.uk/3.pdf>

Particles and Nuclei 2

- 2/24/26
- Announcements:
- HW 3 Due
- Reading: For next time → Griffiths the rest of Chapter 6
Lancaster Blundell Chunk (Propagator)

CP Violation – Matter/AM Asymm

- Antimatter vs Matter distinguisher

$$(a) \pi^+ + e^- + \bar{\nu}_e \quad \text{or} \quad (b) \pi^- + e^+ + \nu_e$$

$$\frac{BR(K_L^0 \rightarrow \pi^- e^+ \nu_e) - BR(K_L^0 \rightarrow \pi^+ e^- \bar{\nu}_e)}{BR(K_L^0 \rightarrow \pi^- e^+ \nu_e) + BR(K_L^0 \rightarrow \pi^+ e^- \bar{\nu}_e)} = .0033$$

Notice –
an “A”
form

- Matter/Antimatter Asymmetry of Universe ?
- CP Violation not big → complicated ways to explain
 - have imagined ways cause neither are other parts of the Big Bang understood (e.g, Inflation)
- Heavy Ion Collisions – Asymm generated by the QGP?
 - Originally one minor goal of checking – QGP results in total symmetry matter antimatter (correcting for original matter baryons)
- Regions of Antimatter—more Anti matter?
 - AMS Experiment

Total Conservation

	Q	L	B	J	P	G	C	CP	T	CPT
Strong	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Electromagnetic	✓	✓	✓	✓	✓	X	✓	✓	✓	✓
Weak	✓	✓	✓	✓	X	X	X	X	X	✓

Q = electric charge, L = lepton number, B = baryon number, J = spin,
P = parity, G = G-parity, C = charge conjugation, T = time reversal

CP, CPT are products

Not included....quark flavour...which is conserved by the strong and electromagnetic interaction and by the neutral weak interaction but not by the charged weak interaction. Also, could add isospin I and third component of isospin I_3 (also, did not include cons. of energy etc).

We will discuss CP, T, and CPT next time.